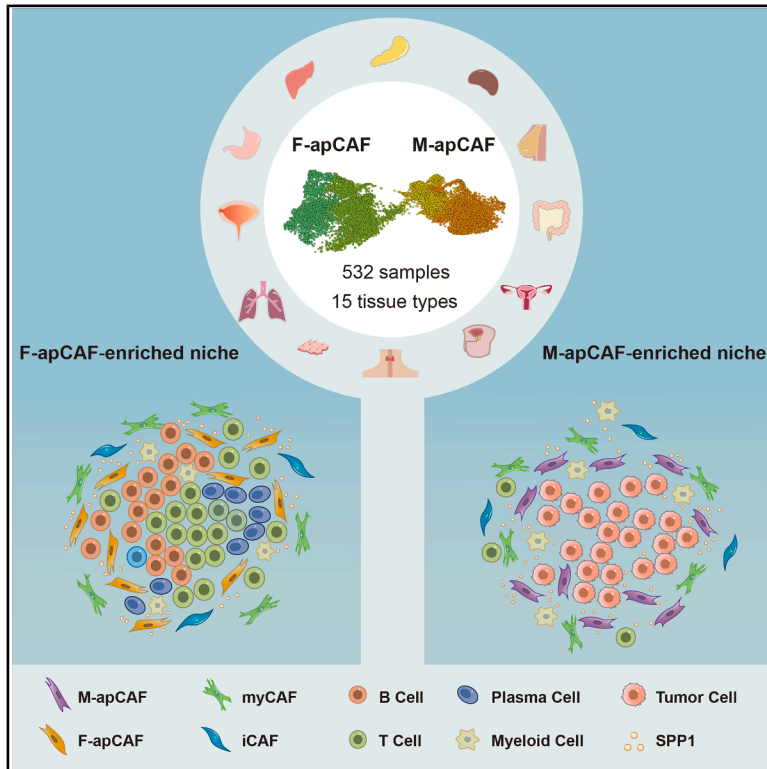


Single-cell resolution spatial analysis of antigen-presenting cancer-associated fibroblast niches

Graphical abstract



Authors

Xiongfeng Chen, Zhuan Zhou, Luyu Xie, ..., Herbert J. Zeh 3rd, Alex C. Kim, Huocong Huang

Correspondence

alex.kim@utsouthwestern.edu (A.C.K.), huocong.huang@utsouthwestern.edu (H.H.)

In brief

Chen et al. identify two distinct antigen-presenting cancer-associated fibroblast (apCAF) populations across multiple cancer types. Mesothelial-related apCAFs locate near cancer cells while fibrocyte-related apCAFs associate with lymphocyte-enriched niches. Both populations highly express SPP1, which facilitates tumor growth, metastasis, desmoplasia, and therapeutic resistance.

Highlights

- Two distinct apCAF populations identified across cancers
- M-apCAFs locate near cancer cells; F-apCAFs associate with lymphocyte-enriched niches
- Both apCAF subtypes highly express SPP1, driving tumor progression and metastasis



Article

Single-cell resolution spatial analysis of antigen-presenting cancer-associated fibroblast niches

Xiongfeng Chen,^{1,2} Zhuan Zhou,^{1,2} Luyu Xie,³ Kailiang Qiao,⁴ Yiyue Jia,^{1,2} Shunheng Liu,² Zeynep Yazgan,² Francesca Rossi,^{1,6} Yang Liu,⁵ Bo Zhang,⁵ Patricio M. Polanco,¹ Herbert J. Zeh 3rd,¹ Alex C. Kim,^{1,2,*} and Huocong Huang^{1,2,5,7,*}

¹Department of Surgery, University of Texas Southwestern Medical Center, Dallas, TX 75390, USA

²Hamon Center for Therapeutic Oncology Research, University of Texas Southwestern Medical Center, Dallas, TX 75390, USA

³Department of Health Data Science and Biostatistics, University of Texas Southwestern Medical Center, Dallas, TX 75390, USA

⁴Children's Medical Center Research Institute, University of Texas Southwestern Medical Center, Dallas, TX 75390, USA

⁵Department of Immunology, University of Texas Southwestern Medical Center, Dallas, TX 75390, USA

⁶Present address: Immunology Program, The Babraham Institute, Babraham Research Campus, Cambridge, CB22 3AT, UK

⁷Lead contact

*Correspondence: alex.kim@utsouthwestern.edu (A.C.K.), hucong.huang@utsouthwestern.edu (H.H.)

<https://doi.org/10.1016/j.ccell.2025.09.001>

SUMMARY

Recent studies identify a unique subtype of cancer-associated fibroblasts (CAFs) termed antigen-presenting CAFs (apCAFs), which remain poorly understood. To gain a comprehensive understanding of the origin and function of apCAFs, we construct a fibroblast molecular atlas across 15 types of tissues and solid tumors. Our integration study unexpectedly reveals two distinct apCAF populations present in most cancer types: one associated with mesothelial-like cells and the other with fibrocytes. Using a high-resolution single-cell spatial imaging platform, we characterize the spatial niches of these apCAF populations. We find that mesothelial-like apCAFs are located near cancer cells, while fibrocyte-like apCAFs are associated with lymphocyte-enriched niches. Additionally, we discovered that both apCAF populations can up-regulate secreted phosphoprotein 1 (SPP1), which facilitates primary tumor formation, peritoneal metastasis, and therapy resistance. Taken together, this study offers an unprecedented resolution in analyzing apCAFs and their spatial niches.

INTRODUCTION

Fibroblasts are mesenchymal cells essential for preserving tissue integrity, regulating inflammatory responses and fibrosis, and facilitating wound repair.^{1–3} Cancer is often described as a wound that never heals, making fibroblasts major components of the tumor microenvironment.⁴ In this context, they perform various functions, including the synthesis and modification of the extracellular matrix, complex signaling with cancer cells, and interactions with infiltrating immune cells.³ Cancer-associated fibroblasts (CAFs) have thus become attractive targets for improving cancer therapies. However, the diversity and plasticity of CAFs present significant challenges in developing effective CAF-targeted treatments.^{5–8} Single-cell RNA sequencing (scRNA-seq) techniques significantly advance our understanding of fibroblast heterogeneity, revealing distinct fibroblast subtypes with specific functions in both healthy and diseased tissues.² Many CAF populations are identified across different cancer types.^{9–15} However, the lineage relationships between CAF subtypes remain poorly understood due to the limited num-

ber of cross-tissue comparison studies and inconsistencies in CAF nomenclature.⁸

Emerging evidence suggests that phenotypically, CAFs can be classified into three major subtypes based on their molecular signatures: inflammatory CAFs (iCAFs), myofibroblastic CAFs (myCAFs), and antigen-presenting CAFs (apCAFs).¹⁰ This classification system provides a consistent framework to study the lineage and function of CAFs. Characterized by the expression of human leukocyte antigen class II (HLA II, human) or major histocompatibility complex class II (MHC II, mouse) molecules, apCAFs can effectively present antigens and regulate T cells.^{10,16} Our previous study also found that apCAFs are derived from mesothelial progenitors in murine pancreatic ductal adenocarcinoma (PDAC).¹⁶ However, due to their tissue- or spatial-dependent abundance, cellular plasticity, and mixed molecular signatures of immune cells and fibroblasts, apCAFs remain the least understood and characterized CAF subtype.

Major questions regarding apCAFs include their existence in human tumors, their origins in different cancer types, and the spatial niches they occupy. To address these questions, we



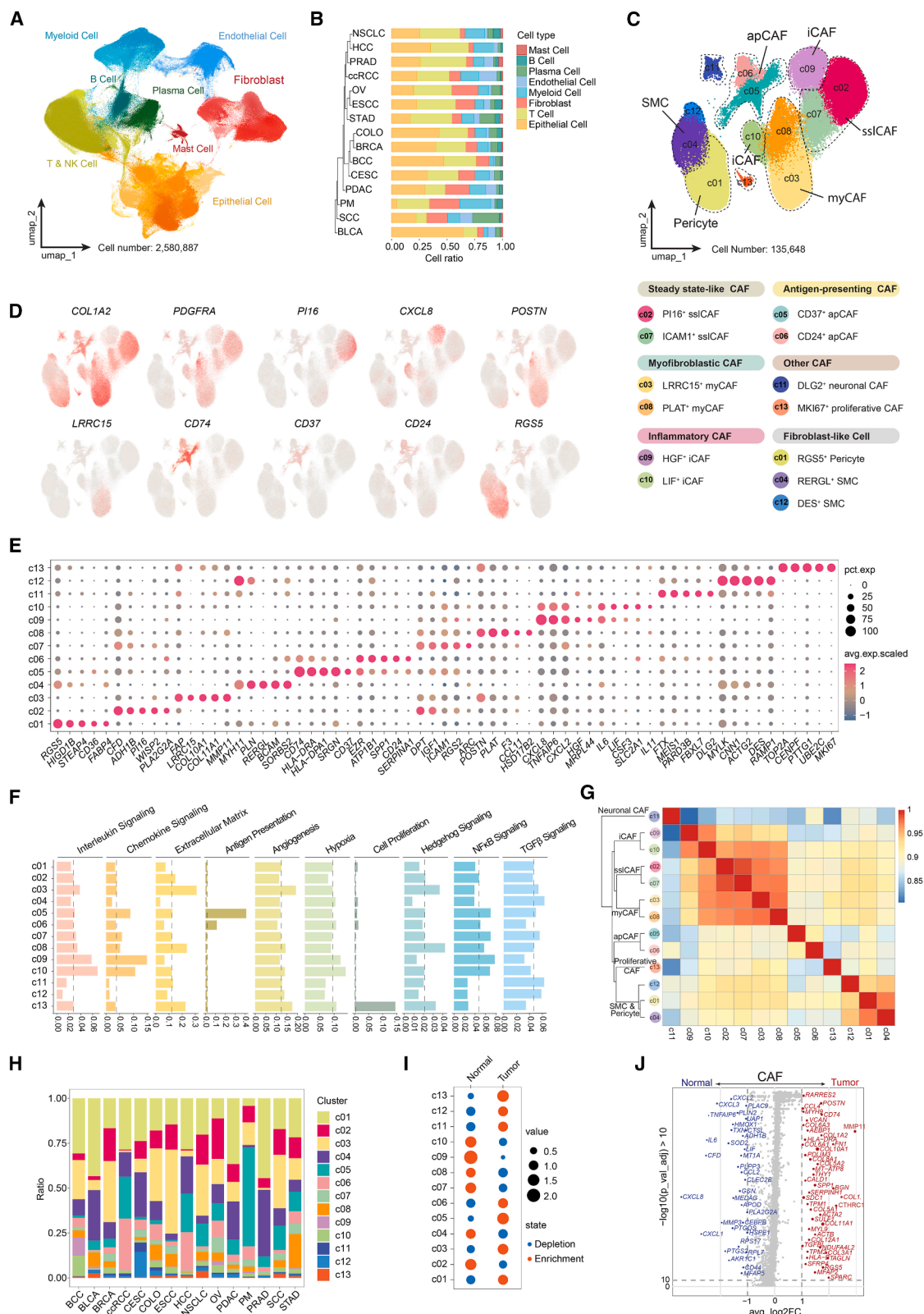


Figure 1. Cross-tissue comparison of CAF subtypes
(A) UMAP of integrated scRNA-seq atlas across 15 tissue and tumor types.
(B) Major cell type proportions by cancer type.

integrate scRNA-seq data from 532 human samples, encompassing over 2.5 million cells, to construct a comprehensive fibroblast molecular atlas across 15 types of tissues and solid tumors. Our robust integration study surprisingly reveals two distinct populations of apCAFs that exist in most tissue types, with a high abundance in specific tissues such as the peritoneum, kidney, and liver. One population is associated with mesothelial-like cells, as we previously reported, while the other is linked to fibrocytes.^{16,17}

To understand the stromal niches formed by the two apCAF populations, we spatially deconvolve the CAF subtype signatures derived from the atlas using a high-resolution single-cell spatial imaging platform. This approach reveals that mesothelial-like apCAFs are proximal to cancer cells, whereas fibrocyte-like apCAFs are closely associated with lymphocyte-enriched regions. Additionally, we discovered that apCAFs are the major stromal cells that express secreted phosphoprotein 1 (SPP1), which facilitates tumor formation, metastases, therapy resistance, and myCAF formation. This work provides an analysis of apCAFs and niches at an unprecedented resolution, advancing our understanding of this poorly understood CAF subtype.

RESULTS

Cancer-associated fibroblast phenotypes are conserved across cancer types

To comprehensively characterize apCAF in human cancer, we first collected scRNA-seq datasets derived from human tumors and the associated normal tissues across 13 different organs including bladder (BLCA), breast (BRCA), cervix (CESC), colorectum (COLO), esophagus (ESCC), kidney (ccRCC), liver (HCC), lung (NSCLC), ovary (OV), pancreas (PDAC), peritoneum (PM), prostate (PRAD), skin (BCC/SCC) and stomach (STAD) (Figure S1A; Table S1). We performed rigorous quality control by filtering cells with <200 or >6000 detected genes and those with >15% mitochondrial unique molecular identifier (UMI) count percentage. Genes detected in <3 cells were removed, and potential doublets were excluded. After batch effect correction and integrative clustering, we generated a high-quality scRNA-seq atlas comprising 532 samples and >2.5 million cells (Figures 1A and S1B). Major cell types were identified by distinct signatures: epithelial cells, T & NK cells, fibroblast-like cells, myeloid cells, endothelial cells, plasma cells, B cells, and mast cells (Figures 1A and S1C). PDAC, PM, and OV showed relatively higher fibroblast-like cell

proportions than other cancer types (Figure 1B), though this may reflect variations in tissue processing protocols across studies. We extracted fibroblast-like cells using marker genes *COL1A1*, *COL1A2*, and *COL3A1*, revealing 13 distinct clusters (Figures 1C–1F and S1D; Table S2). While all clusters expressed *COL1A2*, clusters 01, 04, and 12 (c01, 04, 12) lacked pan-fibroblast markers *PDGFRA* and *LUM* (Figures 1D and S1D). Instead, c01 expressed pericyte markers (*RGS5*, *MCAM*, *CSPG4*), while c04 and c12 showed smooth muscle cell (SMC) markers (*MYH11*, *PLN*) (Figures 1D, 1E, and S1D). The remaining clusters represented fibroblasts across tissues.

The fibroblasts formed 6 major subtypes based on their signatures and signaling: iCAFs (c09, c10), myCAFs (c03, c08), apCAFs (c05, c06), steady-state-like CAFs (ssICAFs) (c02, c07), neuronal CAFs (c11), and proliferative CAFs (c13). iCAFs expressed inflammatory genes (*CXCL2*, *CXCL8*, *IL6*) with enriched interleukin/chemokine/NFκB signaling and hypoxia signature (Figures 1D–1F; Table S3). myCAFs expressed *POSTN* and *MMP11* with extracellular matrix and hedgehog signaling (c03 marked by *LRR15*, c08 by *PLAT*) (Figures 1D–1F and S1D). In contrast, we found that common activated CAF markers *ACTA2* and *FAP* were not myCAF-specific (Figure S1D). Two robust apCAF clusters (c05, c06) were found, which featured HLA II genes (*CD74*, *HLA-DRA*, *HLA-DPA1*) and antigen presentation function, with c05 expressing *SRGN*, *CD37*, and *C1Q* genes, while c06 expressed *CD24* and *SERPINA1* (Figure 1D, 1E, and S1D). ssICAFs expressed universal resident fibroblast markers *DPT* and *CD34*, with c02 also expressing progenitor marker *PI16*, serving as possible progenitors for iCAFs and myCAFs (Figures 1D, 1E, and S1D).² The two minor clusters c11 showed neuronal signatures (*DGL2*, *PIEZO2*), while c13 was proliferative (*MKI67*) (Figures 1E and S1D). TGFβ signaling was highest in c03 myCAFs, c07 ssICAFs, c11 neuronal CAFs, and SMCs (Figure 1F).

Hierarchical clustering showed ssICAFs, iCAFs, and myCAFs belonged to one branch while apCAFs formed a distinct branch, suggesting different progenitor origins (Figure 1G). Despite proportional differences, CAF subtypes were surprisingly conserved across cancer types, with one exception being HGF-high c09 iCAFs, which were enriched in skin and ovarian cancers (Figure 1H). Comparing the relative percentage alteration of each fibroblast cluster between normal tissues and tumors, *LRR15*+ myCAFs (c03), both apCAF subtypes (c05, c06), neuronal CAFs, and proliferative CAFs increased in tumors, while steady-state like (c02, c07) and inflammatory fibroblasts

(C) UMAP of fibroblast clusters shows 13 CAF subpopulations classified into ssICAF, myCAF, iCAF, apCAF, neuronal CAF, proliferative CAF, and fibroblast-like cells (pericyte, SMC).

(D) UMAP of key marker genes: *COL1A2*, *PDGFRA*, *PI16*, *CXCL8*, *POSTN*, *LRR15*, *CD74*, *CD37*, *CD24*, and *RGS5*.

(E) Bubble heatmap shows expression patterns of signature genes in fibroblast populations. Dot size: fraction of expressing cells; color: Z score of scaled expression.

(F) Molecular signature scores across CAF clusters. Dashed lines: median scores.

(G) Heatmap shows pairwise Pearson correlations among CAF subsets, with highly similar CAF subsets clustering into adjacent modules. Color bar: correlation coefficients.

(H) CAF population percentages by cancer type.

(I) Cellular abundance of fibroblast subclusters in normal and tumor tissues. Normalized values > 1: enrichment, <1: depletion. Dot size: normalized value; color: enrichment category.

(J) All CAF subsets from the scRNA-seq data are grouped into normal and tumor conditions. Differential expression between normal and tumor CAF subsets is visualized by a volcano plot.

Also see Figure S1; Tables S1, S2, and S3.

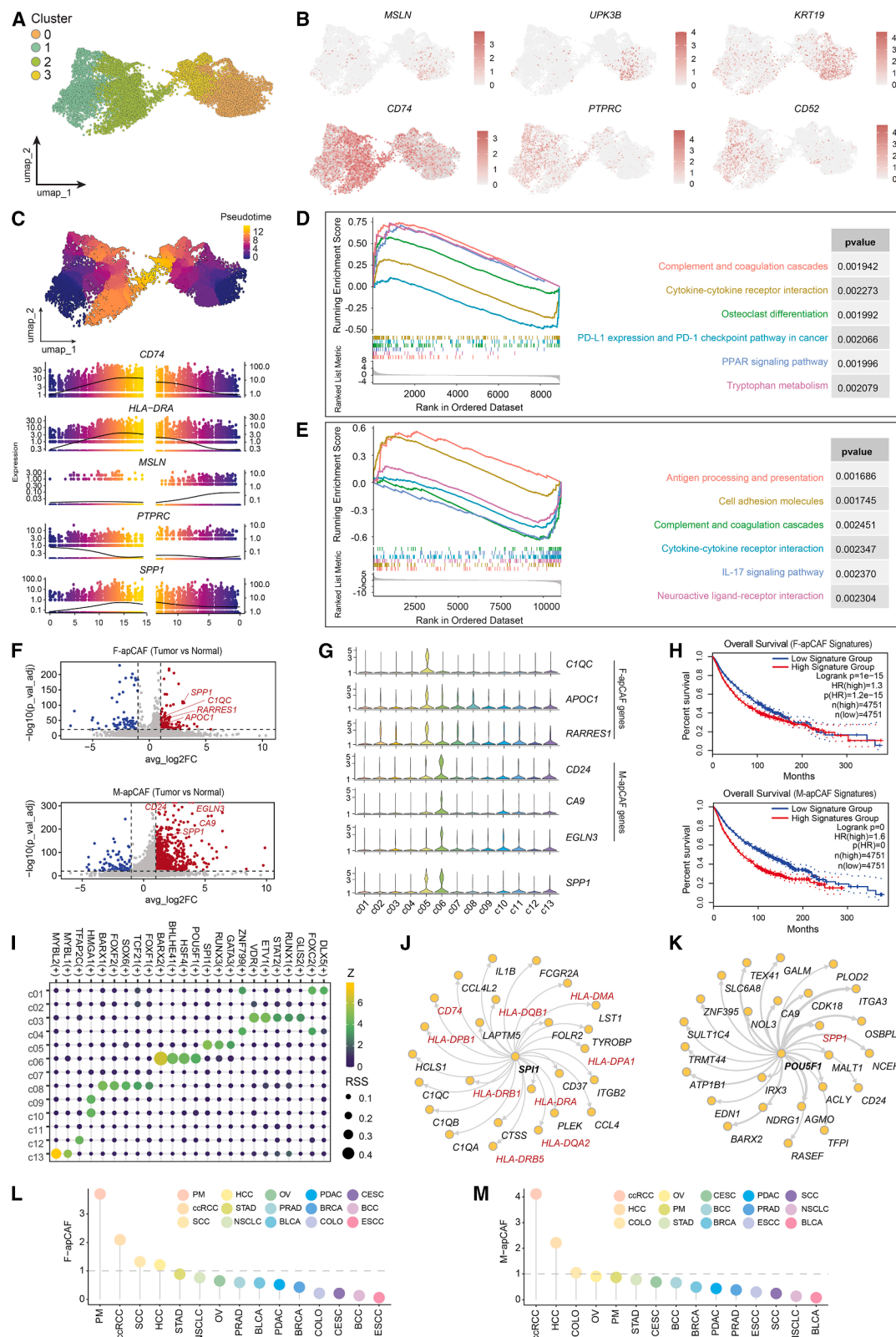


Figure 2. Two distinct populations of apCAFs

(A) UMAP of all apCAFs (HLA II+) extracted from Figure 1C, revealing four subclusters after re-clustering.
(B) UMAP of signature genes: *CD74*, *MSLN*, *UPK3B*, *KRT19*, *PTPRC*, *CD52*.

(legend continued on next page)

(c09, c10) relatively decreased (Figure 1I). Consistently, in tumors, fibroblasts globally showed increased myofibroblastic and antigen-presenting signatures with decreased inflammatory signature compared to normal tissue fibroblasts (Figure 1J).

Identification of two distinct populations of antigen-presenting cancer-associated fibroblasts

To further interrogate the origin and molecular features of the two apCAF populations, we extracted c05 and c06 from the atlas and performed re-clustering of the cells, which revealed two distinct populations consisting of 4 subclusters (Figures 2A and S1E). One population (clusters 0, 3) carried a mesothelial-like signature marked by mesothelial genes (*MSLN*, *UPK3B*, *KRT19*) (Figure 2B).¹⁶ Moreover, this population also expressed *CD24*, a recently reported innate immune checkpoint and involved in the modulation of cell stemness and differentiation (Figure S1F).¹⁸ These data are consistent with our previous study demonstrating the mesothelial origin of apCAFs. In comparison, we discovered the other apCAF population (clusters 1, 2) carried fibrocyte features (Figures 2A and 2B). Fibrocytes are reported to be derived from a hematopoietic origin and express hematopoietic markers such as *CD45* and fibroblastic markers such as extracellular matrix proteins at the same time.^{17,19–21} They are known to be able to migrate to wound sites and facilitate inflammation and fibrotic response. We found the fibrocyte-like apCAFs were marked by many hematopoietic genes such as *PTPRC* (gene for *CD45*), *CD37*, and *CD52* (Figures 2B and S1F). Therefore, in this study, we named the mesothelial-associated population M-apCAFs and the fibrocyte-associated population F-apCAFs. We then performed pseudotime analysis, which revealed two distinct transitional dynamic trajectories among apCAF populations (Figure 2C). We found that the down-regulation of the mesothelial gene *MSLN* in M-apCAFs, and the down-regulation of *PTPRC* in F-apCAFs, accompanied an up-regulation of HLA II genes (*CD74*, *HLA-DRA*) along the trajectories. Pathway analysis of F-apCAFs (Figure 2D) and M-apCAFs (Figure 2E) suggested that both apCAF populations were heavily involved in immune-regulating activities. However, we found that there was a lack of expression of major costimulatory molecules such as *CD40*, *CD80*, and *CD86* in the two apCAF populations (Figure S1G). These data align with our prior murine PDAC model findings that apCAFs present antigens but

lack costimulatory molecules. This deficiency leads to regulatory T cell induction.¹⁶

We compared differentially expressed genes between normal and cancer tissues within the two apCAF populations. Genes up-regulated in cancer were identified, and the specificity of each gene was assessed by evaluating whether it was distinctly expressed in each apCAF population compared to other CAF subtypes. This analysis led to the identification of three genes most specific to F-apCAFs (*C1QC*, *APOC1*, *RARRES1*) and M-apCAFs (*CD24*, *CA9*, *EGLN3*) (Figures 2F and 2G). The three-gene signatures for both F-apCAFs and M-apCAFs were significantly associated with poor survival outcomes in patients with cancer (Figure 2H). In addition, we found that *SPP1* was commonly up-regulated in both apCAF populations. Importantly, apCAFs exhibited substantially higher expression of *SPP1* compared to other CAF clusters (Figure 2G). Furthermore, *SPP1* expression increased along the pseudotime trajectories of both apCAF populations (Figure 2C). Elevated *SPP1* expression also served as a strong negative prognostic marker for patient survival (Figure S1H).

To understand the regulatory network in apCAFs, we performed SCENIC analysis, which revealed distinct transcription factors that drove the gene program for F-apCAFs (*SPI1*, *RUNX3*, *GATA3*) and M-apCAFs (*BARX2*, *BHLHE41*, *HSF4*, *POU5F1*) (Figures 2I, S1I, and S1J).²² These transcription factors might directly regulate the transcription of the key apCAF signatures. For example, we found *SPI1* could be potential regulator of HLA II genes (*CD74*, *HLA-DRA*, *HLA-DRB1*, *HLA-DRB5*, *HLA-DPA1*, *HLA-DPB1*, *HLA-DQA2*, *HLA-DQB1*, *HLA-DMA*) in F-apCAFs (Figure 2J). We also found *POU5F1* (gene for *OCT4*, the major transcription factor that regulates stemness) might regulate *SPP1* in M-apCAFs (Figure 2K).²³ We then analyzed the abundance of the two apCAF populations across cancer types, and found F-apCAFs were most enriched in peritoneal metastasis (PM), ccRCC, SCC, HCC, and STAD, while M-apCAFs were most enriched in ccRCC, HCC, COLO, OV, and PM (Figures 2L and 2M). Therefore, in the following studies, we chose PM as a key model to further understand apCAFs.

Stromal heterogeneity of peritoneal metastasis

Peritoneum is a complex tissue that consists of numerous stromal cells, including mesothelial cells, fibroblasts, endothelial

(C) Pseudotime trajectories (Monocle3) showing two distinct transitional trajectories among apCAF subgroups with the dynamic expression of *CD74*, *HLA-DRA*, *MSLN*, *PTPRC*, and *SPP1*.

(D) GSEA for F-apCAF subclusters (2 vs. 1). $p < 0.01$ is considered significant.

(E) GSEA for M-apCAF subclusters (3 vs. 0). $p < 0.01$ is considered significant.

(F) Differentially expressed genes between normal and cancer tissues in both apCAF populations (adjusted $p < 1 \times 10^{-20}$ (Wilcoxon rank-sum test) and $\log_2|FC| > 0.75$ are color-coded). Genes up-regulated in cancer are identified, and the specificity of each gene is assessed by evaluating whether it is distinctly expressed in each apCAF population compared to other CAF subtypes. This analysis leads to the identification of three genes most specific to F-apCAFs (*C1QC*, *APOC1*, *RARRES1*) and M-apCAFs (*CD24*, *CA9*, *EGLN3*).

(G) Violin plots show the expression of *C1QC*, *APOC1*, *RARRES1*, *CD24*, *CA9*, *EGLN3*, and *SPP1* across CAF subclusters.

(H) Kaplan-Meier plots of combined overall survival in TCGA solid tumors (all tumor types available on GEPIA2), stratified by F-apCAF (*C1QC*, *APOC1*, *RARRES1*) or M-apCAF (*CD24*, *CA9*, *EGLN3*) signature genes. Data are analyzed using log rank test.

(I) Regulon activity scores for transcription factors in CAF subsets (pySCENIC). Dot size: signal strength; color: Z score.

(J) Predicted *SPI1* transcription factor regulatory network in F-apCAFs. HLA II genes highlighted.

(K) Predicted *POU5F1* transcription factor regulatory network in M-apCAFs. *SPP1* highlighted.

(L) F-apCAF relative ratios across cancer types. Enrichment is defined as values >1 (over-representation), with a horizontal baseline for comparative assessment.

(M) M-apCAF relative ratios across cancer types. Enrichment is defined as values >1 (over-representation), with a horizontal baseline for comparative assessment.

Also see Figure S1.

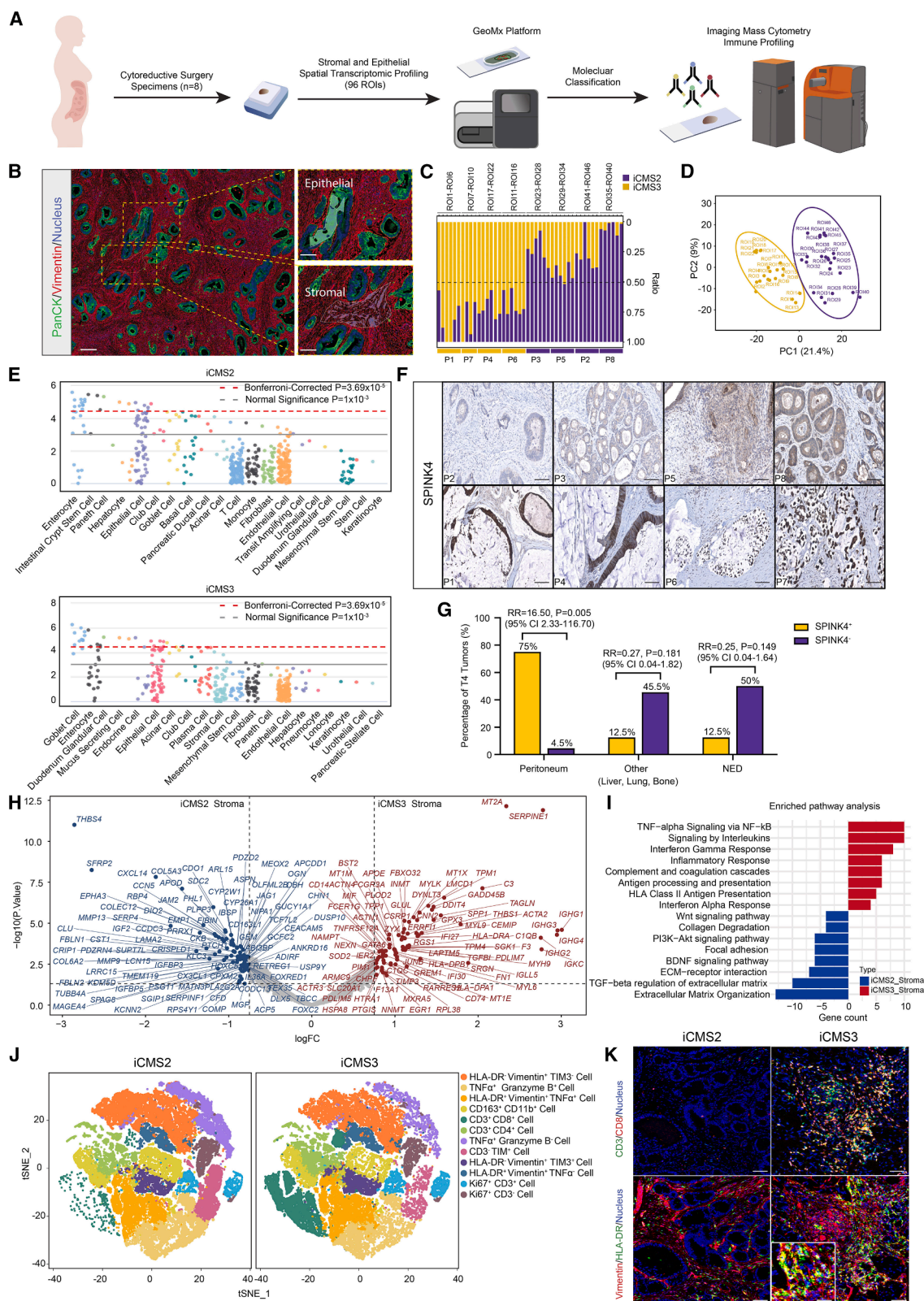


Figure 3. Molecular classification of PM stroma

(A) PM molecular classification workflow using GeoMx spatial transcriptomics and imaging mass cytometry.

(B) Representative morphology marker staining (pan-cytokeratin (PanCK), vimentin, nuclei) from GeoMx assay. Scale: 100µm; inset: 50µm.

(legend continued on next page)

cells, and immune cells.²⁴ This specific microenvironment may facilitate the formation of apCAFs during metastasis. One of the most common origins of PM is colon cancer. About 20% of patients with colon cancer develop PM. Cancer cells from primary colon cancer can be classified into two intrinsic epithelial phenotypes (iCMS2 and iCMS3) based on common epithelial developmental programs.¹⁴ Therefore, it is likely that PM can also be derived from two different cancer cell lineages and consists of heterogeneous stroma. Before specifically looking into apCAFs, we first tried to have a global understanding of PM epithelial and stromal heterogeneity using an unbiased whole-genome spatial transcriptomic approach.

We collected tissue specimens from 8 patients with PM originating from colon cancer. Using Nanostring GeoMx whole transcriptome atlas spatial RNA profiling, we characterized molecular signatures of tumor epithelium and adjacent stroma (Figures 3A and 3B). Deconvolution and principal component analysis of epithelial regions of interest (ROIs) signatures revealed two molecular subtypes: iCMS2- or iCMS3-like (Figures 3C and 3D; Table S4). Cell-type specific enrichment showed iCMS2 epithelial cells resembled enterocytes while iCMS3 epithelial cells resembled goblet cells, indicating two cell lineage programs within PM cancer cells (Figure 3E). To assess their prognostic significance, we examined primary colon cancer cases at T4 stage using IHC staining with iCMS3 marker SPINK4 (Figures 3F, S2A, and S2B). We found a significant association between SPINK4⁺ invasion fronts and elevated PM risk compared to SPINK4⁻ counterparts (relative risk = 16.5, $p = 0.005$) (Figure 3G), underscoring iCMS3 epithelial cells as potential drivers of PM.

Next, we analyzed the transcriptomic information of tumor cell-adjacent stroma. When we grouped stromal ROIs into iCMS2 and iCMS3 categories based on deconvolution analysis, we found iCMS3 stroma displayed enrichment in HLA II antigen presentation and inflammation-related pathways (Figures 3H and 3I; Table S5). Considering the unique inflammatory profiles observed in iCMS3 PM, we speculated potential differences in immune cell infiltration between iCMS2 PM and iCMS3 PM. To explore this, we employed imaging mass cytometry in both subtypes. Our analysis suggested a trend toward increased T cells and HLA II⁺Vimentin⁺ stromal cells within iCMS3 PM (Figures 3J, 3K, and S2C–S2E). We validated changes of T cells in a larger cohort of PM samples (Figures S2F and S2G), supporting iCMS3 PM is more immunologically active.

Spatial analysis of F-antigen-presenting cancer-associated fibroblast niches

The HLA II⁺ stromal signature in iCMS3 PM may be partially attributed to the presence of apCAFs. Although the GeoMx platform provides unbiased whole-genome transcriptomic signatures, the cellular resolution of the technique is limited. To further dissect the spatial distribution of apCAFs and the associated niches at a single cell resolution, we exploited the Xenium *In Situ* high-plex spatial imaging platform. We designed a fully customized panel of probes consisting of 480 genes based on the scRNA-seq atlas we generated (Figure 1; Table S6). We then performed the Xenium assays with 8 human PM samples of colon origin, including one PM-adjacent tissue. We obtained a total of 799,455 high quality cells across samples. Unsupervised clustering analysis resulted in the identification of major cell types, including CAFs, SMCs, endothelial cells, lymphocytes, myeloid cells, iCMS2, and iCMS3 cancer cells (Figures 4A and 4B). Among the 8 PM samples, 3 samples carried a distinct iCMS3 epithelial signature (PM1, PM2, PM3), 2 carried an iCMS2 signature (PM6, PM7), while the other 2 samples had a mix of both iCMS2 and iCMS3 signatures (PM4, PM5) (Figure S3A). We found that the presence of iCMS3 cancer cells was associated with an increased infiltration of lymphocytes (Figures 4C and S3A), consistent with the stromal signatures revealed by the GeoMx assay (Figure 3).

Next, we conducted the robust cell type decomposition to deconvolve spatial transcriptomic data into cell types using the reference constructed from our pan-cancer scRNA-seq atlas (Figure 4D) and then cellular niche composition analysis (Figure 4E). We found that in the iCMS3 PM samples with marked lymphocyte infiltration, four distinct spatial niches could be identified: vasculature-, lymphocyte-, stroma-, and tumor-enriched niches. Niche composition analysis revealed that myCAFs and myeloid cells were the predominant stromal populations in stroma- and tumor-enriched niches, while vasculature-enriched niches primarily consisted of endothelial cells and pericytes. Remarkably, F-apCAFs represented the most abundant cell type within lymphocyte-enriched niches (Figure 4E). To validate this spatial association, we selected ten lymphocyte-enriched niches across PM samples containing iCMS3 cancer cells and quantified the proportions of each cell type (Figures 4F, 4G, and S3B). This analysis confirmed that F-apCAFs were the most prevalent CAF subtype in lymphocyte-enriched regions (Figure 4G).

Given the co-expression of hematopoietic and fibroblast markers by F-apCAFs, we performed analyses to confirm their

(C) Deconvolution of GeoMx spatial RNA data with iCMS2/iCMS3 signatures. Epithelial ROIs can be classified as iCMS2-like (patient (P)2, 3, 5, 8) or iCMS3-like (patient (P)1, 4, 6, 7).

(D) PCA of all ROIs in iCMS2 and iCMS3 PM groups.

(E) Cell-type enrichment analysis using differentially expressed genes from iCMS2 and iCMS3 PM.

(F) SPINK4 IHC in iCMS2 (P2, 3, 5, 8) and iCMS3 (P1, 4, 6, 7) PM groups. Scale: 50μm.

(G) PM and other site metastasis percentages and relative risks in T4 colon cancer with SPINK4^{+/+} invasion borders. Relative risks (RR) computed using bivariate log-binomial regressions. CI: confidence interval; NED: no evidence of disease.

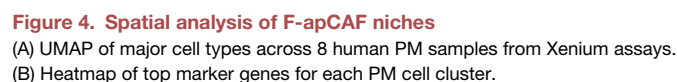
(H) Differentially expressed genes between iCMS2 and iCMS3 PM stroma ($p < 0.05$ (Wilcoxon rank-sum test), $\log_2[FC] > 0.75$).

(I) Pathway enrichment based on differentially expressed genes between iCMS2 and iCMS3 PM stroma.

(J) Imaging mass cytometry in iCMS2 and iCMS3 PM samples ($n = 3/\text{group}$, 2 images/sample) with 20 immune markers. t-SNE plots of major cell subclusters by iCMS subtype.

(K) Representative images showing T cells (CD3, CD8) and HLA II⁺ stromal cells (HLA-DR, vimentin). Yellow: CD3⁺CD8⁺ or HLA II⁺Vimentin⁺ cells. Scale: 50μm; inset: 20μm.

Also see Figure S2 and Tables S4 and S5.



Cancer Cell 43, 2224–2240, December 8, 2025 2231

distinct cellular identity. First, we compared F-apCAF gene signatures with those of major immune cell populations (macrophages, monocytes, dendritic cells, B cells, plasma cells, and T cells) extracted from our pan-cancer cell atlas. While F-apCAFs expressed the hematopoietic marker *PTPRC* (CD45), they were clearly distinguished by the robust expression of fibroblast-specific genes, including *COL1A2*, *COL1A1*, *COL3A1*, and *PDGFRA*, which were absent in immune cell populations (Figure S3C). To specifically address whether F-apCAFs might represent follicular dendritic cells (FDCs), which can be present within lymphoid structures, we extracted FDC marker genes from published scRNA-seq studies of splenic fibroblastic cells (Figure S3D).²⁵ We found that neither F-apCAFs nor M-apCAFs expressed FDC markers, including *CXCL13*, *SPIB*, *CR2*, and *CCL19* (Figure S3E), definitively distinguishing them from FDCs. Finally, we validated the presence of PDGFR α ⁺HLA-DRA⁺CD45⁺ F-apCAFs in lymphocyte-enriched niches through multiplex IHC (Figure S3F). Collectively, these analyses establish F-apCAFs as a distinct fibrocyte-like cell population that combines hematopoietic lineage markers with functional fibroblast characteristics.

Although myCAFs also contributed to a major cell proportion in the lymphocyte-enriched niches, we found F-apCAFs had higher density and closer interaction with lymphocyte aggregates compared to myCAFs (average distance 37.86 μ m vs. 66.06 μ m) (Figure 4H). To investigate the interaction between F-apCAFs and immune cells, we first identified the myeloid cell and lymphocyte subtypes from our pan-cancer scRNA-seq atlas (Figures S4A–S4D).^{26–29} We then performed ligand-receptor interaction analysis with F-apCAF-derived ligands and immune cell-expressing receptors. The results revealed many important signals that mediated the immune-regulating function of F-apCAFs (Figures 4I and 4J). For example, *CCL5*, which transcribes a chemoattractant for T cells and monocytes, probably contributes to the lymphocyte recruitment.³⁰ However, immunosuppressive genes were also found, such as *SPP1* and *LAI1*, suggesting the complex role of F-apCAFs in inflamed tumors.^{31,32}

Spatial analysis of M-antigen-presenting cancer-associated fibroblast niches

Compared to F-apCAFs, M-apCAFs showed more context-dependent spatial distribution (Figure S4E). Since the peritoneum is lined by mesothelial cells, cancer metastasis to this site may induce M-apCAF formation through direct mesothelial cell interaction.^{16,24} M-apCAFs express cytokeratin genes (Figure 2B), and IHC staining revealed cytokeratin⁺ CAFs infil-

trating PM stroma, typically located near cytokeratin⁺ mesothelium (Figures 5A and S5A). Multiplex IHC with cancer cell marker (SPINK4) and CAF marker (PDGFR α) confirmed these CAFs were not cancer cell-derived (Figures 5A and S5B). In samples with marked cytokeratin⁺ CAF infiltration, spatial transcriptomic cell type deconvolution and niche composition analysis identified distinct mesothelial-enriched niches alongside stroma-, vasculature- and tumor-enriched niches (Figures 5B and 5C). These mesothelial niches contained M-apCAFs (23.89%), myeloid cells (15.48%), and T cells (12.38%) as major stromal components (Figure 5C). Some mesothelial-enriched niches were continuous with normal mesothelium (marked by *MSLN*, *UPK3B*) (Figure S5C), suggesting mesothelial-to-M-apCAF differentiation. Most mesothelial-enriched niches were adjacent to tumor epithelium (Figure 5D), indicating M-apCAFs may closely interact with immune and cancer cells.

Our prior study demonstrated that M-apCAFs were able to induce the formation of immunosuppressive T cells in murine models.¹⁶ Consistently, we found T cells within the mesothelial-enriched niches expressed a high level of suppressive markers (*TIGIT*, *CTLA4*, *LAG3*, *PDCD1*, *ICOS*, *FOXP3*) (Figure 5E). Further ligand-receptor interaction analysis suggested that *SPP1* was a major ligand that mediated the interaction between M-apCAFs and immune cells or cancer cells (Figures S5D and S5E).

Knockout of host secreted phosphoprotein 1 results in a reduction of peritoneum and ascites formation

SPP1, also known as osteopontin, is a highly phosphorylated secreted protein that normally functions as an extracellular structural protein in bone. In the wound and cancer settings, SPP1 is up-regulated as an acute phase protein.³³ It is associated with inflammation and fibrotic reaction.^{34–36} In cancer, SPP1 is known to exert multifaceted tumor-promoting functions by promoting cellular plasticity and chemoresistance in cancer cells and serving as an immune checkpoint in T cells, but the effect of SPP1 in PM is unknown.^{31,37} Interestingly, we observed that iCMS3 PM tends to have a higher level of stroma-derived SPP1 (Figure S5F), potentially contributing to the peritoneal tropism of this PM subtype. SPP1 can be expressed by cancer cells, myeloid cells, and CAFs.^{38,39} In fact, a prior study in breast cancer has reported apCAFs as an important SPP1 producer.⁴⁰ Moreover, our computational data suggest apCAFs is a major source of SPP1 among other CAF subtypes, we validated this result by examining the expression of total and secreted SPP1 in cancer cells (EpCAM⁺), myeloid cells (CD11b⁺), apCAFs (PDGFR α ⁺MHC II⁺) and other CAFs (PDGFR α ⁺MHC II[−]) isolated

(C) Major cell type proportions in non-tumor adjacent (NA), iCMS2, and iCMS3 PM samples.

(D) Cell types in representative iCMS3 PM sample, deconvolved using RCTD algorithm with scRNA-seq atlas as reference. Dashed regions magnified in panel F. Scale: 1000 μ m.

(E) Left: Spatial niches computed by the k-nearest neighbor algorithm. Right: Cell type composition across four distinct spatial niches. Scale: 1000 μ m.

(F) Cellular spatial distribution in four representative lymphocyte-enriched niches. Scale: 100 μ m.

(G) Proportional distribution of cell types across 10 lymphocyte-enriched niches from 5 PM samples (1–3 niches/sample). Each colored line: a specific lymphocyte-enriched niche; thick line within box: mean proportion of each cell type; dashed line: overall mean proportion.

(H) Cell density and spatial distance analysis of F-apCAFs or myCAFs from lymphocyte-enriched aggregates.

(I) Estimated probability of ligand-receptor pairs from F-apCAFs (ligands) to immune cells (receptors). P-values are calculated by comparing the inferred communication probability with those in permuted sets, with $p < 0.05$ considered significant.

(J) Significant ligand-receptor pairs ($p < 0.05$) between F-apCAF-derived ligands and immune cell receptors.

Also see Figures S3 and S4 and Table S6.

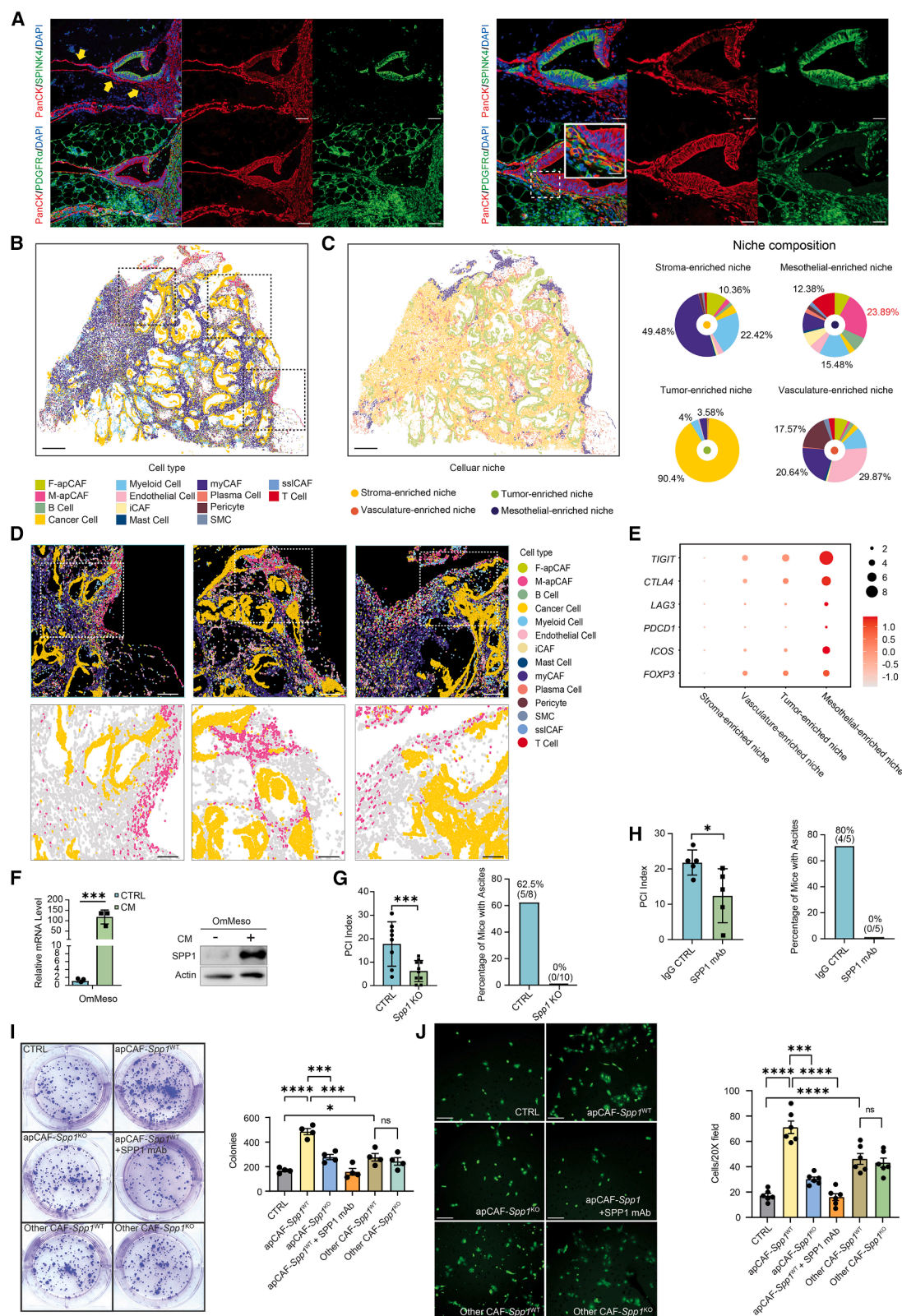


Figure 5. Spatial analysis of M-apCAF niches

(A) Multiplex IHC of human PM tissue showing PanCK, SPINK4, PDGFR α , and DAPI. Arrows: PanCK⁺ mesothelial cells. Scale: 50 μ m (left), 25 μ m (right), 6 μ m (inset).

(legend continued on next page)

from *in vivo* PM nodules established from MC38 murine colon cancer cells. The data showed that apCAFs produced the highest level of SPP1 compared to other stromal cells (Figures S5G–S5I). Additionally, we treated the mesothelial cells isolated from normal mouse omentum (OmMeso cells) with tumor conditioned media and observed a marked elevation of SPP1 expression in OmMeso cells (Figure 5F).

To test the effect of SPP1 on cancer cell migration and invasion, we performed scratch and Matrigel invasion assays with MC38 cells. Adding SPP1 protein significantly enhanced, while anti-SPP1 monoclonal antibody (mAb) inhibited cancer cell migration and invasion (Figures S5J and S5K). We then investigated the role of host SPP1 in PM formation by injecting MC38 cells intraperitoneally into wildtype or *Spp1* knockout mice, finding that host *Spp1* knockout substantially reduced PM and ascites formation (Figure 5G). This phenotype was recapitulated by treating wildtype PM-bearing mice with anti-SPP1 mAb (Figure 5H). To assess the importance of apCAF-derived SPP1, we established PM nodules in wildtype or *Spp1* knockout mice using MC38 cells. Isolated apCAFs (PDGFR α ⁺MHC II⁺) and other CAFs (PDGFR α ⁺MHC II[−]) were cocultured with MC38 cells for colony formation and Matrigel invasion assays. SPP1 secreted by apCAFs significantly increased colony-forming and invasion capabilities of MC38, which anti-SPP1 mAb inhibited (Figures 5I and 5J). Together, these data suggest stromal SPP1 might drive PM formation, with apCAFs producing SPP1 to facilitate this process.

M-antigen-presenting cancer-associated fibroblast are associated with secreted phosphoprotein 1-positive niches in pancreatic ductal adenocarcinoma

Given the previous studies of apCAFs in PDAC, we also examined the PDAC model.^{10,16} We performed Xenium assays in 3 treatment-naïve and 3 chemoradiotherapy-treated human PDAC samples. Unsupervised clustering analysis identified 6 major cell types including CAFs, endothelial cells, lymphocytes, myeloid cells, and two populations of cancer cells (cancer cell 1 and cancer cell 2) (Figures S6A and S6B). We found that after the chemoradiotherapy treatment, tissues became more desmoplastic with a larger CAF proportion, consistent with a recent report (Figure S6C).⁴¹ Among the six samples, we found three had more robust T&B cell signatures (Figure S6D). After cellular

deconvolution, we found F-apCAFs were also associated with lymphocyte-enriched areas in PDAC (Figures 6A–6C). We further validated this result in a larger cohort of human PDAC samples by multiplex IHC. Out of 30 patients, 8 exhibited lymphocyte-enriched areas (CD3⁺CD19⁺). We quantified PDGFR α ⁺HLA-DRA⁺CD45⁺ F-apCAFs in 14 regions across these 8 samples, comparing lymphocyte-enriched versus non-enriched areas. F-apCAFs were significantly more abundant in lymphocyte-enriched areas (Figures 6D and 6E).

After chemoradiotherapy, the cancer cell 1 population decreased while the cancer cell 2 proportion remained stable, indicating cancer cell 2 was therapy-resistant (Figure S6C). Differential gene analysis revealed cancer cell 2 was characterized by high *SPP1* expression (Figure S6E), creating SPP1⁺ and SPP1⁺ tumor epithelium across the tissue. Intriguingly, M-apCAFs were enriched in SPP1⁺ tumor areas (Figures 6F and 6G), which also contained infiltrating SPP1⁺ myeloid cells that represent protumor polarization (Figure 6F).³⁸ These findings suggest M-apCAFs contribute to forming an SPP1⁺ niche that promotes tumor progression and therapy resistance in PDAC.

To confirm apCAFs as a major SPP1 source in PDAC, we treated the pancreatic mesothelial cell line (PanMeso cells) with PDAC organoid-conditioned media, which up-regulated SPP1 expression (Figure 6H). We then tested *in vivo* induction by orthotopically co-injecting GFP-labeled PanMeso cells with PDAC cell lines (BMFA3 or CT1BA5). Flow-sorted GFP⁺ PanMeso cells from established tumors showed dramatically increased *Spp1* expression compared to parental cells (Figure 6I). We then investigated the effect of host SPP1 on PDAC progression by orthotopically injecting a syngeneic PDAC cell line (6620c1) into wildtype or *Spp1*-knockout mice, and found significantly reduced tumor growth with host *Spp1* depletion (Figure 6J). Anti-SPP1 mAb treatment recapitulated these results (Figure 6K). To assess the specific impact of apCAF-derived SPP1, we isolated PDGFR α ⁺MHC II⁺ apCAFs from wildtype or *Spp1*-knockout tumors for coculture experiments. SPP1 from apCAFs significantly enhanced 6620c1 colony formation and chemoresistance (Figures 6L and 6M). Similarly, PanMeso cell coculture increased 6620c1 chemoresistance, an effect abolished by *Spp1* knockdown or anti-SPP1 mAb (Figures 6N and 6F). These findings demonstrate that apCAF-derived SPP1 directly promotes PDAC progression and chemoresistance.

(B) Cell types in representative PM sample, deconvolved using RCTD algorithm with the scRNA-seq atlas as reference. Dashed regions magnified in panel D. Scale: 800 μ m.

(C) Left: Spatial niches computed by the k-nearest neighbor algorithm. Right: Cell type composition across four spatial niches. Scale: 800 μ m.

(D) Cellular spatial distribution in three M-apCAF-enriched regions. Scale: 200 μ m; inset: 100 μ m.

(E) T cell immunosuppressive gene expression across four spatial niches.

(F) SPP1 expression in OmMeso cells after tumor conditioned medium (CM) treatment by RT-PCR ($n = 3$ /group) and western blot. Mean $\pm$ SEM; *** $p < 0.001$ (t-test).

(G) Measurement of peritoneal cancer index (PCI) scores and ascites in wildtype (WT, $n = 8$) or *Spp1* knockout (KO, $n = 10$) mice with intraperitoneal MC38 cells for 4 weeks. Mean $\pm$ SEM; *** $p < 0.001$ (t-test).

(H) Measurement of PCI scores and ascites in mice with intraperitoneal MC38 cells for 4 weeks treated with control Ab or anti-SPP1 mAb ($n = 5$ /group, mice treated with Ab 1 week after cancer cell injection and maintained at 2 doses/week). Mean $\pm$ SEM; * $p < 0.05$ (t-test).

(I) Colony formation of MC38 cells cocultured with apCAFs (PDGFR α ⁺MHC II⁺) or other CAFs (PDGFR α ⁺MHC II[−]) isolated from WT or *Spp1* KO PM nodules. Effect of anti-SPP1 mAb is also tested in the WT apCAF coculture. CTRL: cancer monoculture; $n = 4$ /group. Mean $\pm$ SEM; * $p < 0.05$, *** $p < 0.001$, **** $p < 0.0001$ (t-test), ns: not significant.

(J) Invasion of CFSE-labeled MC38 cells in Matrigel transwells mixed with apCAFs (PDGFR α ⁺MHC II⁺) or other CAFs (PDGFR α ⁺MHC II[−]) isolated from WT or *Spp1* KO PM nodules (1:1). Effect of anti-SPP1 mAb is also tested in the WT apCAF group. CTRL: cancer alone; $n = 6$ /group. Mean $\pm$ SEM; *** $p < 0.001$, **** $p < 0.0001$ (t-test), ns: not significant. Scale: 25 μ m.

Also see Figures S4 and S5.

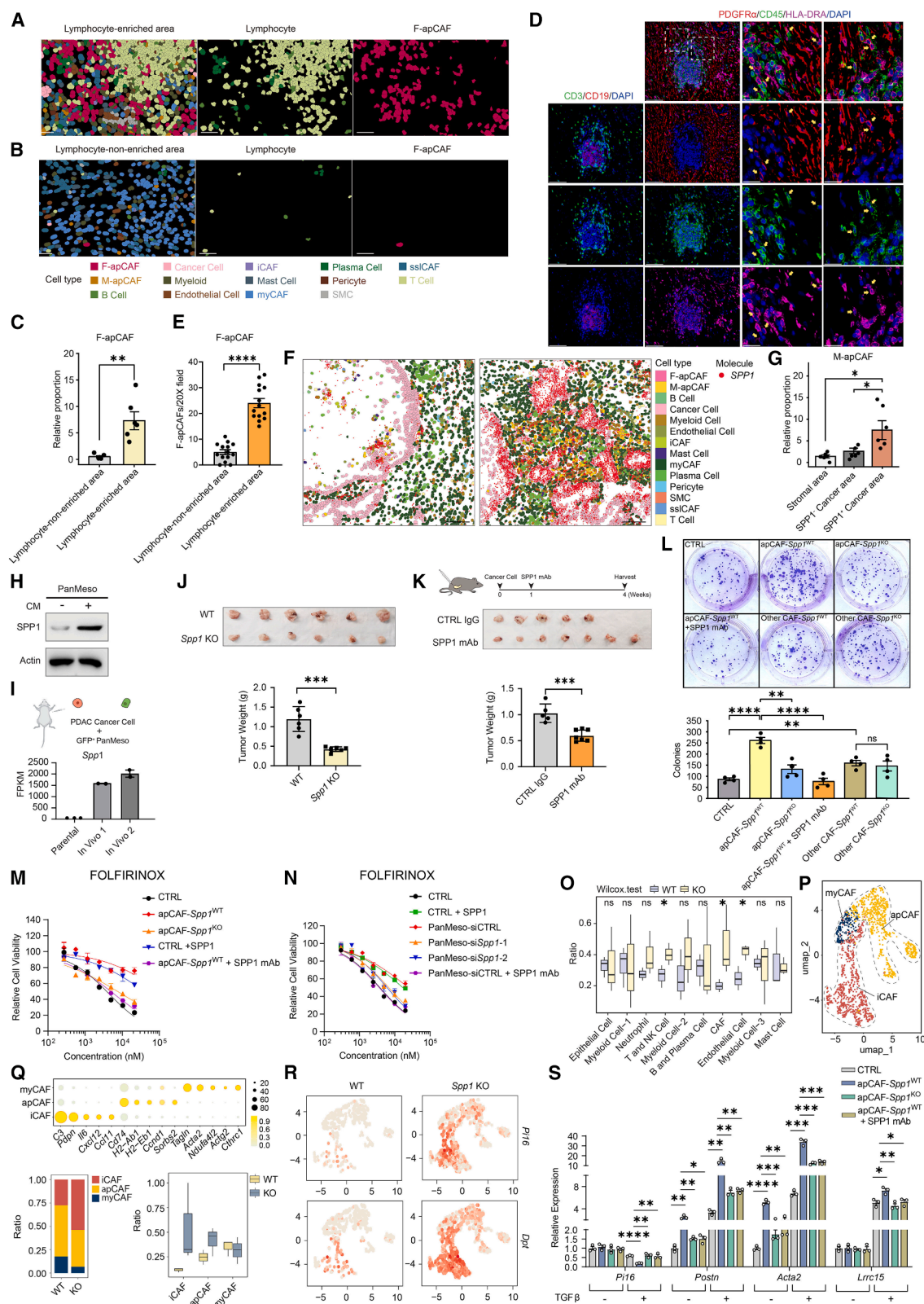


Figure 6. apCAFs contribute to SPP1⁺ tumor-promoting niches

(A) Cell types in lymphocyte-enriched niches in PDAC are shown after deconvolution using the RCTD algorithm. Lymphocytes and F-apCAFs are also shown in separate panels, respectively. Scale: 10μm.

(legend continued on next page)

Antigen-presenting cancer-associated fibroblast-derived secreted phosphoprotein 1 facilitates fibrotic barrier formation in pancreatic ductal adenocarcinoma

To understand the microenvironmental effect of host *Spp1* knockout, we performed scRNA-seq on tumors from control and knockout groups (Figures S6G and S6H). Results revealed significantly increased T cell infiltration in host *Spp1* knockout tumors (Figure 6O), which we confirmed by IHC staining (Figure S6I). However, *Spp1* knockout also increased fibroblast (Figure 6O), suggesting possible phenotypic alterations in fibroblasts promoting T cell infiltration. Unsupervised re-clustering of fibroblasts from both groups identified iCAFs, myCAFs and apCAFs (Figures 6P and 6Q), with *Spp1* expression primarily enriched in apCAFs (Figures S6H and S6J). We validated these findings by examining total and secreted SPP1 in different cell populations (EpCAM⁺ cancer cells, CD11b⁺ myeloid cells, PDGFRα⁺MHC II⁺ apCAFs, and PDGFRα⁺MHC II⁻ other CAFs) isolated from 6620c1 orthotopic tumors, confirming apCAFs as the major stromal SPP1 source (Figures S6K and S6L).

When comparing CAF subtype proportions after *Spp1* knockout, we observed reduced myCAFs with increased iCAFs and apCAFs (Figure 6Q). The apCAF expansion likely represents a compensatory mechanism attempting to restore SPP1 signaling. Importantly, CAFs in the knockout group up-regulated sslCAF markers (*Pi16*, *Dpt*) (Figure 6R), suggesting normalization toward steady-state features found in normal tissues. CytoTRACE analysis supported iCAFs/sslCAFs as myCAF pro-

genitors (Figure S6M).⁴² Additionally, T cell chemoattractant signatures were up-regulated in *Spp1* knockout CAFs (Figure S6N), potentially explaining increased T cell infiltration. We validated these findings by treating pancreatic stellate cells with TGFβ, which is a major signal for myCAF differentiation,^{43,44} in the presence of wildtype or *Spp1*-knockout apCAFs, demonstrating that apCAF-derived SPP1 significantly promoted myCAF differentiation, shown by decreased progenitor gene (*Pi16*) expression and increased mature myCAF markers (*Postn*, *Acta2*, *Lrrc15*) (Figure 6S). Together, these results establish apCAFs as a major stromal SPP1 source facilitating desmoplastic barrier formation.

DISCUSSION

CAFs have traditionally been viewed as a homogeneous population within the tumor microenvironment. However, advances in single-cell transcriptomic and proteomic technologies have enabled researchers to profile CAF heterogeneity across various cancer types.^{9,10,13,45–47} While numerous CAF subtypes have been identified, the lack of consensus on nomenclature and subtype markers has obscured the relationships between findings from different studies.⁸ This underscores the urgent need for comprehensive cross-tissue comparisons to establish a unified framework for understanding CAFs. Our study confirms that iCAFs, myCAFs, and apCAFs exist universally across tumor types. Moreover, we identified another significant CAF phenotype carrying universal fibroblast progenitor signatures (marked

(B) Cell types in lymphocyte-non-enriched niches in PDAC are shown after deconvolution using the RCTD algorithm. Lymphocytes and F-apCAFs are also shown in separate panels, respectively. Scale: 10μm.

(C) Proportions of F-apCAFs from the Xenium assay are quantified in lymphocyte-enriched and -non-enriched niches across PDAC samples (*n* = 6/niche). Mean ± SEM; ***p* < 0.01 (t-test).

(D) Multiplex IHC of human PDAC tissue showing CD3, CD19, PDGFRα, HLA-DRA, CD45, and DAPI. Arrows: PDGFRα⁺HLA-DRA⁺CD45⁺ F-apCAFs. Scale: 25μm, 3μm (inset).

(E) Proportions of F-apCAFs from the multiplex IHC are quantified in lymphocyte-enriched and -non-enriched niches across PDAC samples (*n* = 14/niche). Mean ± SEM; *****p* < 0.0001 (t-test).

(F) Cellular spatial distribution and *SPP1* expression in *SPP1*⁻ and *SPP1*⁺ cancer areas. Scale: 100μm.

(G) Proportions of M-apCAFs from the Xenium assay are quantified in stromal, *SPP1*⁻, and *SPP1*⁺ cancer areas (*n* = 6/area) across PDAC samples. Mean ± SEM; **p* < 0.05 (t-test).

(H) *SPP1* expression in PanMeso cells after PDAC tumor CM treatment by western blot.

(I) GFP⁺ PanMeso cells are co-injected with murine PDAC cells (BMFA3: *In Vivo* 1 or CT1BA5: *In Vivo* 2; 1:1 ratio) to form tumors. GFP⁺ cells are then flow-sorted and analyzed by RNA-seq for *Spp1* expression versus parental PanMeso cells. Mean ± SEM (parental: *n* = 3; *In Vivo* 1 and 2: *n* = 2).

(J) Tumor weight after orthotopic injection of 6620c1 PDAC cells in WT or *Spp1* KO mice (*n* = 6/group). Mean ± SEM; ****p* < 0.001 (t-test).

(K) Tumor weight after orthotopic injection of 6620c1 cells in mice treated with control Ab (*n* = 5) or anti-SPP1 mAb (*n* = 7). Mean ± SEM; ****p* < 0.001 (t-test).

(L) Colony formation of 6620c1 cells cocultured with apCAFs (PDGFRα⁺MHC II⁺) or other CAFs (PDGFRα⁺MHC II⁻) isolated from WT or *Spp1* KO PDAC tumors. Effect of anti-SPP1 mAb is also tested in the WT apCAF coculture. CTRL: cancer monoculture; *n* = 4/group. Mean ± SEM; ***p* < 0.01, *****p* < 0.0001 (t-test), ns: not significant.

(M) Viability of CFSE-labeled 6620c1 cells mixed with apCAFs (PDGFRα⁺MHC II⁺) isolated from WT or *Spp1* KO PDAC tumors (1:1) after FOLFIRINOX treatment. Effect of recombinant SPP1 and anti-SPP1 mAb is also tested in the CTRL or WT apCAF group, respectively. CTRL: cancer cell alone; *n* = 3/condition. Mean ± SEM.

(N) Viability of CFSE-labeled 6620c1 cells mixed with PanMeso cells (1:1) transfected with control (siCTRL) or two independent siRNAs against *Spp1* (si*Spp1*-1 and si*Spp1*-2) after FOLFIRINOX treatment. Effect of recombinant SPP1 and anti-SPP1 mAb is also tested in the CTRL or siCTRL group, respectively. CTRL: cancer cell alone; *n* = 3/condition. Mean ± SEM.

(O) Cell type ratios between WT and *Spp1* KO tumors analyzed by scRNA-seq (6 tumors/group, pooled as 3 libraries/group). Range of ratios of each cell type is shown. Thick line within box: mean proportion of each cell type. **p* < 0.05 (Wilcoxon test), ns: not significant.

(P) iCAF, myCAF, and apCAF clusters identified from WT and *Spp1* KO tumor scRNA-seq data.

(Q) Top: Marker gene expression across CAF subsets. Dot size indicates proportion of cells expressing each gene within clusters; color shows normalized expression levels. Bottom: Proportional changes of CAF subtypes between WT and *Spp1* KO groups. Range of ratio of each cell type is shown by a boxplot. Line within box: mean proportion of each cell type.

(R) UMAP of sslCAF markers *Pi16* and *Dpt* expression in WT and *Spp1* KO tumors.

(S) RT-PCR measuring gene expression changes (*Pi16*, *Postn*, *Acta2*, *Lrrc15*) in PSCs cocultured with *Spp1* WT or KO apCAFs with or without TGFβ. Effect of anti-SPP1 mAb is also tested in the WT apCAF group. CTRL: PSC monoculture; *n* = 3/group. Mean ± SEM; **p* < 0.05, ***p* < 0.01, ****p* < 0.001, *****p* < 0.0001 (t-test). Also see Figure S6.

by *DPT, PI16, CD34*) as reported by Buechler et al.² This population contributes substantially to the tumor stroma, potentially associates with less aggressive tumor regions.⁴⁸

Our group and Elyada et al. first identified apCAFs in PDAC through scRNA-seq analysis in murine models.^{9,10} These cells are defined by their expression of MHC II genes and have subsequently been detected across multiple cancer types.^{41,49–51} However, debate persists regarding whether apCAFs constitute a genuine CAF population, and conflicting functions of apCAFs have been reported, given their relative scarcity compared to myCAFs in human, context-dependent nature, and incomplete understanding of their cellular origins.^{51–57} To address these knowledge gaps, we employed a big data cross-tissue approach combined with spatial transcriptomics at single-cell resolution. Our findings confirm that apCAFs are present in most solid tumors and, surprisingly, originate from two distinct populations. The first population, which we term M-apCAFs, is associated with mesothelial cells as we previously reported.¹⁶ Mesothelial cells possess both epithelial and mesenchymal characteristics with plasticity under specific conditions.⁵⁸ We found this population expressed cytokeratin genes. We speculate other mesothelial-like epithelial cells can also serve as progenitors for M-apCAFs in a tissue-dependent manner. The second population, termed F-apCAFs, is associated with fibrocytes—bone marrow-derived cells exhibiting signatures of both fibroblasts and hematopoietic cells that can produce extracellular matrix and inflammatory proteins.^{19,20} Despite their low abundance in circulation, fibrocytes potentially promote inflammation and fibrosis by migrating to injury sites.

The proximity of F-apCAFs to lymphocyte-enriched niches suggests they may play a critical role in lymphocyte recruitment and regulation, consistent with the reported association of fibrocytes with inflammatory phenotypes.⁵⁹ Recent studies have begun to recognize the presence of apCAFs within tertiary lymphoid structures (TLS) across various cancer types.^{48,55,60} However, these studies may not have fully appreciated the existence of two distinct apCAF populations due to several methodological limitations: (1) the lack of comprehensive cross-tissue apCAF characterization to define subtype-specific signatures, (2) lower spatial resolution mapping techniques, and (3) limited marker panels that do not capture the full spectrum of apCAF heterogeneity. For instance, Keridani et al. found that apCAFs promote tumor immunity via C1q in lung cancer.⁵⁴ Since our analysis reveals that C1q is predominantly produced by F-apCAFs rather than M-apCAFs, their findings likely reflect F-apCAF-specific activity rather than a general apCAF function. Similarly, other studies reporting apCAF-TLS associations may have primarily captured F-apCAF populations without recognizing the distinct M-apCAF lineage.^{55,60} Our high-resolution spatial analysis and comprehensive lineage characterization provide the necessary framework to dissect these previously unrecognized functional distinctions. While our study focused on lymphocyte-enriched areas, the spatial association between F-apCAFs and lymphocyte aggregates warrants further investigation into their specific role in TLS formation and maintenance. The fibrocyte origin of F-apCAFs, with their inherent migratory capacity and immune regulatory functions, positions them as likely orchestrators of organized lymphocyte recruitment and activation. Our observations suggest that F-apCAFs tend to localize at the periphery of lymphocyte-enriched regions. This peripheral positioning may

be optimal for several key functions: (1) capturing tumor-derived antigens and presenting them to T cells at the immune-stromal interface, (2) establishing chemokine gradients that guide lymphocyte trafficking, and (3) providing paracrine signals that support TLS maintenance and expansion. Future studies employing robust genetic lineage tracing systems, potentially utilizing split-Cre technology, will be essential to definitively validate F-apCAF developmental trajectories and function. Additionally, isolation and functional characterization of F-apCAFs from human tissues will be crucial for understanding their therapeutic potential in cancer immunotherapy.

In contrast, M-apCAFs commonly localize adjacent to cancer cells in a context-dependent manner. Since mesothelial-like cells are less likely to be recruited into tumor stroma from circulation (unlike fibrocytes), M-apCAF formation likely depends on local progenitor cell presence and specific differentiation signals from the tumor microenvironment. For instance, in PM, these niches appear more robust near mesothelial regions where mesothelial-to-fibroblast transition can occur. While myeloid cells and T cells are present in M-apCAF niches, these T cells exhibit exhausted, immunosuppressive phenotypes, suggesting M-apCAFs may drive localized immunosuppression rather than immune activation.

Moreover, we validated apCAFs as a major source of stromal SPP1, aligning with previous breast cancer findings.⁴⁰ Targeting SPP1 represents a promising therapeutic strategy for PM and PDAC, particularly given that humanized SPP1 antibodies are already in clinical trials for other diseases.⁶¹ Therefore, the effects of targeting SPP1 to overcome chemoradiotherapy and immunotherapy resistance should be explored clinically.

In summary, our comprehensive cross-tissue comparison identifies two distinct apCAF populations with different cellular origins and characterizes their spatial niches at unprecedented resolution. These findings significantly advance our understanding of this CAF subtype. Future work should incorporate sophisticated genetic lineage tracing tools to definitively map the developmental trajectories of the proposed apCAF populations, which would further enhance our understanding of the distinct origins and functions of apCAFs in the tumor microenvironment.

Limitations of the study

The small cohort size of peritoneal metastasis samples limits our ability to draw definitive conclusions about the relationship between epithelial subtypes (iCMS2/iCMS3) and specific stromal compositions. While we observed trends suggesting iCMS3 PM exhibits a more inflamed signature, a larger patient cohort would be necessary to establish robust associations and properly stratify this heterogeneous disease. Additionally, the limited sample size prevented a comprehensive analysis of clinical outcomes in relation to these molecular subtypes. Future studies with expanded cohorts and integrated single-cell and spatial methodologies will be essential to validate our findings and consolidate the functional significance of apCAFs in the peritoneal metastatic setting.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Huocong Huang (huocong.huang@utsouthwestern.edu).

Materials availability

All unique reagents generated in this study are available from the [lead contact](#) with a completed Materials Transfer Agreement.

Data and code availability

Details of the published scRNA-seq datasets we used for integrated analysis can be found in the [Table S1](#). The spatial transcriptomic data and scRNA-seq generated in this study have been uploaded to GEO under the following accession numbers GEO: GSE274609, GSE274623, GSE274673. RNA-seq data of PanMeso cells have been previously deposited at GEO: GSE196740. Additional information related to the data in this study is available from the [lead contact](#) upon request.

ACKNOWLEDGMENTS

This work was supported by NIH R00 CA252009 and SSO ACPMP Young Investigator subaward to H.H. We thank the UT Southwestern Core facilities (supported in part by P30 CA142543), including Tissue Management, Microarray/Immune Phenotyping, Histopathology, Flow Cytometry, and Whole Brain Microscopy facilities.

AUTHOR CONTRIBUTIONS

XC: acquisition of data, analysis, interpretation of data, and drafting of the article. ZZ: acquisition of data. LX: acquisition of data. KQ: acquisition of data. YJ: acquisition of data. SL: acquisition of data. ZY: acquisition of data. FR: acquisition of data. YL: acquisition of data. BZ: acquisition of data. PMP: tissue collection. HJZ: study design and interpretation of data. ACK: study concept and design, tissue collection, and interpretation of data. HH: study concept and design, acquisition of data, analysis, interpretation of data, and drafting of the article.

DECLARATION OF INTERESTS

The authors declare no competing interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- **KEY RESOURCES TABLE**
- **EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS**
 - Cell lines
 - Animals
 - Patient samples
- **METHOD DETAILS**
 - Plasmids
 - Immunohistochemistry
 - RT-PCR
 - Western blot analysis
 - Tumor conditioned medium
 - Wound healing assay
 - Isolation of cancer cells, myeloid cells, apCAFs and other CAFs
 - Colony formation assay
 - Matrigel invasion assay
 - Chemosensitivity assay
 - SPP1 ELISA
 - PSC isolation and coculture
 - *In vivo* models
 - Single cell cDNA library preparation and sequencing
 - Single-cell RNA-seq data processing
 - Cell cluster proportion and hierarchical clustering of different cancer types
 - Calculation of signature score
 - Similarity analysis of CAF subclusters
 - Enrichment of fibroblast subclusters in normal and tumor tissues
 - Pseudotime trajectory inference

- Gene regulatory network inference
- Cellular abundance analysis of apCAF
- Cell-cell interaction analysis
- Survival analysis
- Enrichment analysis
- CytoTRACE analysis
- GeoMx spatial profiling
- GeoMx data analysis
- Imaging mass cytometry
- Xenium *in situ* high-plex assay
- Xenium data processing

● QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.ccell.2025.09.001>.

Received: August 6, 2024

Revised: July 14, 2025

Accepted: September 3, 2025

Published: September 25, 2025

REFERENCES

1. Younesi, F.S., Miller, A.E., Barker, T.H., Rossi, F.M.V., and Hinz, B. (2024). Fibroblast and myofibroblast activation in normal tissue repair and fibrosis. *Nat. Rev. Mol. Cell Biol.* 25, 617–638. <https://doi.org/10.1038/s41580-024-00716-0>.
2. Buechler, M.B., Pradhan, R.N., Krishnamurthy, A.T., Cox, C., Calviello, A.K., Wang, A.W., Yang, Y.A., Tam, L., Caothien, R., Roose-Girma, M., et al. (2021). Cross-tissue organization of the fibroblast lineage. *Nature* 593, 575–579. <https://doi.org/10.1038/s41586-021-03549-5>.
3. Plikus, M.V., Wang, X., Sinha, S., Forte, E., Thompson, S.M., Herzog, E.L., Driskell, R.R., Rosenthal, N., Biernaskie, J., and Horsley, V. (2021). Fibroblasts: Origins, definitions, and functions in health and disease. *Cell* 184, 3852–3872. <https://doi.org/10.1016/j.cell.2021.06.024>.
4. Dvorak, H.F. (2015). Tumors: wounds that do not heal-redux. *Cancer Immunol. Res.* 3, 1–11. <https://doi.org/10.1158/2326-6066.CIR-14-0209>.
5. Caligiuri, G., and Tuveson, D.A. (2023). Activated fibroblasts in cancer: Perspectives and challenges. *Cancer Cell* 41, 434–449. <https://doi.org/10.1016/j.ccell.2023.02.015>.
6. Chhabra, Y., and Weeraratna, A.T. (2023). Fibroblasts in cancer: Unity in heterogeneity. *Cell* 186, 1580–1609. <https://doi.org/10.1016/j.cell.2023.03.016>.
7. Helms, E., Onate, M.K., and Sherman, M.H. (2020). Fibroblast Heterogeneity in the Pancreatic Tumor Microenvironment. *Cancer Discov.* 10, 648–656. <https://doi.org/10.1158/2159-8290.CD-19-1353>.
8. Sahai, E., Atsaturuv, I., Cukierman, E., DeNardo, D.G., Egeblad, M., Evans, R.M., Fearon, D., Gretchen, F.R., Hingorani, S.R., Hunter, T., et al. (2020). A framework for advancing our understanding of cancer-associated fibroblasts. *Nat. Rev. Cancer* 20, 174–186. <https://doi.org/10.1038/s41568-019-0238-1>.
9. Hosein, A.N., Huang, H., Wang, Z., Parmar, K., Du, W., Huang, J., Maitra, A., Olson, E., Verma, U., and Brekken, R.A. (2019). Cellular heterogeneity during mouse pancreatic ductal adenocarcinoma progression at single-cell resolution. *JCI Insight* 5, e129212. <https://doi.org/10.1172/jci.insight.129212>.
10. Elyada, E., Bolisetty, M., Laise, P., Flynn, W.F., Courtois, E.T., Burkhart, R. A., Teinor, J.A., Belleau, P., Biffi, G., Lucito, M.S., et al. (2019). Cross-Species Single-Cell Analysis of Pancreatic Ductal Adenocarcinoma Reveals Antigen-Presenting Cancer-Associated Fibroblasts. *Cancer Discov.* 9, 1102–1123. <https://doi.org/10.1158/2159-8290.CD-19-0094>.
11. Bartoschek, M., Oskolkov, N., Bocci, M., Lötvot, J., Larsson, C., Sommarin, M., Madsen, C.D., Lindgren, D., Pekar, G., Karlsson, G., et al. (2018). Spatially and functionally distinct subclasses of breast

- cancer-associated fibroblasts revealed by single cell RNA sequencing. *Nat. Commun.* 9, 5150. <https://doi.org/10.1038/s41467-018-07582-3>.
12. Hu, H., Piotrowska, Z., Hare, P.J., Chen, H., Mulvey, H.E., Mayfield, A., Noeen, S., Kattermann, K., Greenberg, M., Williams, A., et al. (2021). Three subtypes of lung cancer fibroblasts define distinct therapeutic paradigms. *Cancer Cell* 39, 1531–1547.e10. <https://doi.org/10.1016/j.ccell.2021.09.003>.
13. Steele, N.G., Carpenter, E.S., Kemp, S.B., Sirihorachai, V.R., The, S., Delrosario, L., Lazarus, J., Amir, E.A.D., Gunchick, V., Espinoza, C., et al. (2020). Multimodal Mapping of the Tumor and Peripheral Blood Immune Landscape in Human Pancreatic Cancer. *Nat. Cancer* 1, 1097–1112. <https://doi.org/10.1038/s43018-020-00121-4>.
14. Joanito, I., Wirapati, P., Zhao, N., Nawaz, Z., Yeo, G., Lee, F., Eng, C.L.P., Macalinalao, D.C., Kahraman, M., Srinivasan, H., et al. (2022). Single-cell and bulk transcriptome sequencing identifies two epithelial tumor cell states and refines the consensus molecular classification of colorectal cancer. *Nat. Genet.* 54, 963–975. <https://doi.org/10.1038/s41588-022-01100-4>.
15. Foster, D.S., Januszyk, M., Delitto, D., Yost, K.E., Griffin, M., Guo, J., Guardino, N., Delitto, A.E., Chinta, M., Burcham, A.R., et al. (2022). Multicomic analysis reveals conservation of cancer-associated fibroblast phenotypes across species and tissue of origin. *Cancer Cell* 40, 1392–1406.e7. <https://doi.org/10.1016/j.ccell.2022.09.015>.
16. Huang, H., Wang, Z., Zhang, Y., Pradhan, R.N., Ganguly, D., Chandra, R., Murimwa, G., Wright, S., Gu, X., Maddipati, R., et al. (2022). Mesothelial cell-derived antigen-presenting cancer-associated fibroblasts induce expansion of regulatory T cells in pancreatic cancer. *Cancer Cell* 40, 656–673.e7. <https://doi.org/10.1016/j.ccell.2022.04.011>.
17. Schmidt, M., Sun, G., Stacey, M.A., Mori, L., and Mattoli, S. (2003). Identification of circulating fibrocytes as precursors of bronchial myofibroblasts in asthma. *J. Immunol.* 171, 380–389. <https://doi.org/10.4049/jimmunol.171.1.380>.
18. Barkal, A.A., Brewer, R.E., Markovic, M., Kowarsky, M., Barkal, S.A., Zaro, B.W., Krishnan, V., Hatakeyama, J., Dorigo, O., Barkal, L.J., and Weissman, I.L. (2019). CD24 signalling through macrophage Siglec-10 is a target for cancer immunotherapy. *Nature* 572, 392–396. <https://doi.org/10.1038/s41586-019-1456-0>.
19. Phillips, R.J., Burdick, M.D., Hong, K., Lutz, M.A., Murray, L.A., Xue, Y.Y., Belperio, J.A., Keane, M.P., and Strieter, R.M. (2004). Circulating fibrocytes traffic to the lungs in response to CXCL12 and mediate fibrosis. *J. Clin. Investig.* 114, 438–446. <https://doi.org/10.1172/JCI20997>.
20. Direkze, N.C., Hodivala-Dilke, K., Jeffery, R., Hunt, T., Poulsom, R., Oukrif, D., Alison, M.R., and Wright, N.A. (2004). Bone marrow contribution to tumor-associated myofibroblasts and fibroblasts. *Cancer Res.* 64, 8492–8495. <https://doi.org/10.1158/0008-5472.CAN-04-1708>.
21. Han, L., Wu, Y., Fang, K., Sweeney, S., Roesner, U.K., Parrish, M., Patel, K., Walter, T., Piermattei, J., Trimboli, A., et al. (2023). The splanchnic mesenchyme is the tissue of origin for pancreatic fibroblasts during homeostasis and tumorigenesis. *Nat. Commun.* 14, 1. <https://doi.org/10.1038/s41467-022-34464-6>.
22. Aibar, S., González-Blas, C.B., Moerman, T., Huynh-Thu, V.A., Imrichova, H., Hulselmans, G., Rambow, F., Marine, J.C., Geurts, P., Aerts, J., et al. (2017). SCENIC: single-cell regulatory network inference and clustering. *Nat. Methods* 14, 1083–1086. <https://doi.org/10.1038/nmeth.4463>.
23. Takahashi, K., and Yamanaka, S. (2006). Induction of pluripotent stem cells from mouse embryonic and adult fibroblast cultures by defined factors. *Cell* 126, 663–676. <https://doi.org/10.1016/j.cell.2006.07.024>.
24. Cortes-Guiral, D., Hubner, M., Alyami, M., Bhatt, A., Ceelen, W., Glehen, O., Lordick, F., Ramsay, R., Sgarbura, O., Van Der Speeten, K., et al. (2021). Primary and metastatic peritoneal surface malignancies. *Nat. Rev. Dis. Primers* 7, 91. <https://doi.org/10.1038/s41572-021-00326-6>.
25. Cheng, H.W., Onder, L., Novkovic, M., Sonesson, C., Lütge, M., Pikor, N., Scandella, E., Robinson, M.D., Miyazaki, J.I., Tersteegen, A., et al. (2019). Origin and differentiation trajectories of fibroblastic reticular cells in the splenic white pulp. *Nat. Commun.* 10, 1739. <https://doi.org/10.1038/s41467-019-09728-3>.
26. Chu, Y., Dai, E., Li, Y., Han, G., Pei, G., Ingram, D.R., Thakkar, K., Qin, J.J., Dang, M., Le, X., et al. (2023). Pan-cancer T cell atlas links a cellular stress response state to immunotherapy resistance. *Nat. Med.* 29, 1550–1562. <https://doi.org/10.1038/s41591-023-02371-y>.
27. Cheng, S., Li, Z., Gao, R., Xing, B., Gao, Y., Yang, Y., Qin, S., Zhang, L., Ouyang, H., Du, P., et al. (2021). A pan-cancer single-cell transcriptional atlas of tumor infiltrating myeloid cells. *Cell* 184, 792–809.e23. <https://doi.org/10.1016/j.cell.2021.01.010>.
28. Tang, F., Li, J., Qi, L., Liu, D., Bo, Y., Qin, S., Miao, Y., Yu, K., Hou, W., Li, J., et al. (2023). A pan-cancer single-cell panorama of human natural killer cells. *Cell* 186, 4235–4251.e20. <https://doi.org/10.1016/j.cell.2023.07.034>.
29. Andreatta, M., Corria-Osorio, J., Müller, S., Cubas, R., Coukos, G., and Carmona, S.J. (2021). Interpretation of T cell states from single-cell transcriptomics data using reference atlases. *Nat. Commun.* 12, 2965. <https://doi.org/10.1038/s41467-021-23324-4>.
30. Huffman, A.P., Lin, J.H., Kim, S.I., Byrne, K.T., and Vonderheide, R.H. (2020). CCL5 mediates CD40-driven CD4+ T cell tumor infiltration and immunity. *JCI Insight* 5, e137263. <https://doi.org/10.1172/jci.insight.137263>.
31. Klement, J.D., Paschall, A.V., Redd, P.S., Ibrahim, M.L., Lu, C., Yang, D., Celis, E., Abrams, S.I., Ozato, K., and Liu, K. (2018). An osteopontin/CD44 immune checkpoint controls CD8+ T cell activation and tumor immune evasion. *J. Clin. Investig.* 128, 5549–5560. <https://doi.org/10.1172/JCI123360>.
32. Zhang, C.C. (2022). A perspective on LILRBs and LAIR1 as immune checkpoint targets for cancer treatment. *Biochem. Biophys. Res. Commun.* 633, 64–67. <https://doi.org/10.1016/j.bbrc.2022.09.019>.
33. Rangaswami, H., Bulbule, A., and Kundu, G.C. (2006). Osteopontin: role in cell signaling and cancer progression. *Trends Cell Biol.* 16, 79–87. <https://doi.org/10.1016/j.tcb.2005.12.005>.
34. Pardo, A., Gibson, K., Cisneros, J., Richards, T.J., Yang, Y., Becerril, C., Yousem, S., Herrera, I., Ruiz, V., Selman, M., and Kaminski, N. (2005). Up-regulation and profibrotic role of osteopontin in human idiopathic pulmonary fibrosis. *PLoS Med.* 2, e251. <https://doi.org/10.1371/journal.pmed.0020251>.
35. O'Regan, A., and Berman, J.S. (2000). Osteopontin: a key cytokine in cell-mediated and granulomatous inflammation. *Int. J. Exp. Pathol.* 81, 373–390. <https://doi.org/10.1046/j.1365-2613.2000.00163.x>.
36. Lenga, Y., Koh, A., Perera, A.S., McCulloch, C.A., Sodek, J., and Zohar, R. (2008). Osteopontin expression is required for myofibroblast differentiation. *Circ. Res.* 102, 319–327. <https://doi.org/10.1161/CIRCRESAHA.107.160408>.
37. Nallasamy, P., Nimmakayala, R.K., Karmakar, S., Leon, F., Seshacharyulu, P., Lakshmanan, I., Rachagani, S., Mallya, K., Zhang, C., Ly, Q.P., et al. (2021). Pancreatic Tumor Microenvironment Factor Promotes Cancer Stemness via SPP1-CD44 Axis. *Gastroenterology* 161, 1998–2013.e7. <https://doi.org/10.1053/j.gastro.2021.08.023>.
38. Bill, R., Wirapati, P., Messermer, M., Roh, W., Zitti, B., Duval, F., Kiss, M., Park, J.C., Saal, T.M., Hoelzl, J., et al. (2023). CXCL9:SPP1 macrophage polarity identifies a network of cellular programs that control human cancers. *Science* 381, 515–524. <https://doi.org/10.1126/science.ade2292>.
39. Wang, H., Li, N., Liu, Q., Guo, J., Pan, Q., Cheng, B., Xu, J., Dong, B., Yang, G., Yang, B., et al. (2023). Antiandrogen treatment induces stromal cell reprogramming to promote castration resistance in prostate cancer. *Cancer Cell* 41, 1345–1362.e9. <https://doi.org/10.1016/j.ccell.2023.05.016>.
40. Friedman, G., Levi-Galibov, O., David, E., Bornstein, C., Giladi, A., Dadiani, M., Mayo, A., Halperin, C., Pevsner-Fischer, M., Lavon, H., et al. (2020). Cancer-associated fibroblast compositions change with breast cancer progression linking the ratio of S100A4(+) and PDPN(+) CAFs to clinical outcome. *Nat. Cancer* 1, 692–708. <https://doi.org/10.1038/s43018-020-0082-y>.

41. Cui Zhou, D., Jayasinghe, R.G., Chen, S., Herndon, J.M., Iglesia, M.D., Navale, P., Wendl, M.C., Caravan, W., Sato, K., Storrs, E., et al. (2022). Spatially restricted drivers and transitional cell populations cooperate with the microenvironment in untreated and chemo-resistant pancreatic cancer. *Nat. Genet.* 54, 1390–1405. <https://doi.org/10.1038/s41588-022-01157-1>.
42. Gulati, G.S., Sikandar, S.S., Wesche, D.J., Manjunath, A., Bharadwaj, A., Berger, M.J., Ilagan, F., Kuo, A.H., Hsieh, R.W., Cai, S., et al. (2020). Single-cell transcriptional diversity is a hallmark of developmental potential. *Science* 367, 405–411. <https://doi.org/10.1126/science.aax0249>.
43. Krishnamurthy, A.T., Shyer, J.A., Thai, M., Gandham, V., Buechler, M.B., Yang, Y.A., Pradhan, R.N., Wang, A.W., Sanchez, P.L., Qu, Y., et al. (2022). LRRRC15(+) myofibroblasts dictate the stromal setpoint to suppress tumour immunity. *Nature* 611, 148–154. <https://doi.org/10.1038/s41586-022-05272-1>.
44. Biffi, G., Oni, T.E., Spielman, B., Hao, Y., Elyada, E., Park, Y., Preall, J., and Tuveson, D.A. (2019). IL1-Induced JAK/STAT Signaling Is Antagonized by TGFbeta to Shape CAF Heterogeneity in Pancreatic Ductal Adenocarcinoma. *Cancer Discov.* 9, 282–301. <https://doi.org/10.1158/2159-8290.CD-18-0710>.
45. Cords, L., Engler, S., Haberecker, M., Rüschoff, J.H., Moch, H., de Souza, N., and Bodenmiller, B. (2024). Cancer-associated fibroblast phenotypes are associated with patient outcome in non-small cell lung cancer. *Cancer Cell* 42, 396–412.e5. <https://doi.org/10.1016/j.ccell.2023.12.021>.
46. Dominguez, C.X., Müller, S., Keerthivasan, S., Koeppen, H., Hung, J., Gierke, S., Breart, B., Foreman, O., Bainbridge, T.W., Castiglioni, A., et al. (2020). Single-Cell RNA Sequencing Reveals Stromal Evolution into LRRRC15(+) Myofibroblasts as a Determinant of Patient Response to Cancer Immunotherapy. *Cancer Discov.* 10, 232–253. <https://doi.org/10.1158/2159-8290.CD-19-0644>.
47. Hutton, C., Heider, F., Blanco-Gomez, A., Banyard, A., Kononov, A., Zhang, X., Karim, S., Paulus-Hock, V., Watt, D., Steele, N., et al. (2021). Single-cell analysis defines a pancreatic fibroblast lineage that supports anti-tumor immunity. *Cancer Cell* 39, 1227–1244.e20. <https://doi.org/10.1016/j.ccell.2021.06.017>.
48. Gao, Y., Li, J., Cheng, W., Diao, T., Liu, H., Bo, Y., Liu, C., Zhou, W., Chen, M., Zhang, Y., et al. (2024). Cross-tissue human fibroblast atlas reveals myofibroblast subtypes with distinct roles in immune modulation. *Cancer Cell* 42, 1764–1783.e10. <https://doi.org/10.1016/j.ccell.2024.08.020>.
49. Costa, A., Kieffer, Y., Scholer-Dahirel, A., Pelon, F., Bourachot, B., Cardon, M., Sirven, P., Magagna, I., Fuhrmann, L., Bernard, C., et al. (2018). Fibroblast Heterogeneity and Immunosuppressive Environment in Human Breast Cancer. *Cancer Cell* 33, 463–479.e10. <https://doi.org/10.1016/j.ccell.2018.01.011>.
50. Sun, R., Zhao, H., Gao, D.S., Ni, A., Li, H., Chen, L., Lu, X., Chen, K., and Lu, B. (2023). Amphiregulin couples IL1RL1(+) regulatory T cells and cancer-associated fibroblasts to impede antitumor immunity. *Sci. Adv.* 9, eadd7399. <https://doi.org/10.1126/sciadv.add7399>.
51. Varveri, A., Papadopoulou, M., Papadovasilakis, Z., Compeer, E.B., Legaki, A.I., Delis, A., Damaskou, V., Boon, L., Papadogiorgaki, S., Samiotaki, M., et al. (2024). Immunological synapse formation between T regulatory cells and cancer-associated fibroblasts promotes tumour development. *Nat. Commun.* 15, 4988. <https://doi.org/10.1038/s41467-024-49282-1>.
52. Werba, G., Weissinger, D., Kawaler, E.A., Zhao, E., Kalfakakou, D., Dhara, S., Wang, L., Lim, H.B., Oh, G., Jing, X., et al. (2023). Author Correction: Single-cell RNA sequencing reveals the effects of chemotherapy on human pancreatic adenocarcinoma and its tumor microenvironment. *Nat. Commun.* 14, 3912. <https://doi.org/10.1038/s41467-023-39680-2>.
53. Hwang, W.L., Jagadeesh, K.A., Guo, J.A., Hoffman, H.I., Yadollahpour, P., Reeves, J.W., Mohan, R., Drokhyansky, E., Van Wittenberghe, N., Ashenberg, O., et al. (2022). Single-nucleus and spatial transcriptome profiling of pancreatic cancer identifies multicellular dynamics associated with neoadjuvant treatment. *Nat. Genet.* 54, 1178–1191. <https://doi.org/10.1038/s41588-022-01134-8>.
54. Kerdidani, D., Aerakis, E., Verrou, K.M., Angelidis, I., Douka, K., Maniou, M.A., Stamoulis, P., Goudevenou, K., Prados, A., Tzaferis, C., et al. (2022). Lung tumor MHCII immunity depends on *in situ* antigen presentation by fibroblasts. *J. Exp. Med.* 219, e20210815. <https://doi.org/10.1084/jem.20210815>.
55. Song, J., Wei, R., Liu, C., Zhao, Z., Liu, X., Wang, Y., Liu, F., and Liu, X. (2025). Antigen-presenting cancer associated fibroblasts enhance anti-tumor immunity and predict immunotherapy response. *Nat. Commun.* 16, 2175. <https://doi.org/10.1038/s41467-025-57465-7>.
56. Cao, Z., Meng, Z., Li, J., Tian, Y., Lu, L., Wang, A., Huang, J., Wang, J., Sun, J., Chen, L., et al. (2025). Interferon-gamma-stimulated antigen-presenting cancer-associated fibroblasts hinder neoadjuvant chemioimmunotherapy efficacy in lung cancer. *Cell Rep. Med.* 6, 102017. <https://doi.org/10.1016/j.xcrm.2025.102017>.
57. Maru, S.Y., Wetzel, M., Mitchell, J.T., Gross, N.E., Andaloori, L., Howe, K., Kartalia, E., Mo, G., Leatherman, J., Ho, W.J., et al. (2025). Antigen-presenting cancer-associated fibroblasts in murine pancreatic tumors differentially control regulatory T cell phenotype and function via CXCL9 and CCL22. Preprint at bioRxiv. <https://doi.org/10.1101/2025.03.27.645833>.
58. Koopmans, T., and Rinkevich, Y. (2018). Mesothelial to mesenchyme transition as a major developmental and pathological player in trunk organs and their cavities. *Commun. Biol.* 1, 170. <https://doi.org/10.1038/s42003-018-0180-x>.
59. Reilkoff, R.A., Bucala, R., and Herzog, E.L. (2011). Fibrocytes: emerging effector cells in chronic inflammation. *Nat. Rev. Immunol.* 11, 427–435. <https://doi.org/10.1038/nri2990>.
60. Liu, Y., Sinjab, A., Min, J., Han, G., Paradiso, F., Zhang, Y., Wang, R., Pei, G., Dai, Y., Liu, Y., et al. (2025). Conserved spatial subtypes and cellular neighborhoods of cancer-associated fibroblasts revealed by single-cell spatial multi-omics. *Cancer Cell* 43, 905–924.e6. <https://doi.org/10.1016/j.ccell.2025.03.004>.
61. Farrokhi, V., Chabot, J.R., Neubert, H., and Yang, Z. (2018). Assessing the Feasibility of Neutralizing Osteopontin with Various Therapeutic Antibody Modalities. *Sci. Rep.* 8, 7781. <https://doi.org/10.1038/s41598-018-26187-w>.
62. Huang, H., Zhang, Y., Gallegos, V., Sorrelle, N., Zaid, M.M., Toombs, J., Du, W., Wright, S., Hagopian, M., Wang, Z., et al. (2019). Targeting TGFbetaR2-mutant tumors exposes vulnerabilities to stromal TGFbeta blockade in pancreatic cancer. *EMBO Mol. Med.* 11, e10515. <https://doi.org/10.15252/emmm.201910515>.
63. Li, J., Byrne, K.T., Yan, F., Yamazoe, T., Chen, Z., Baslan, T., Richman, L. P., Lin, J.H., Sun, Y.H., Rech, A.J., et al. (2018). Tumor Cell-Intrinsic Factors Underlie Heterogeneity of Immune Cell Infiltration and Response to Immunotherapy. *Immunity* 49, 178–193.e7. <https://doi.org/10.1016/j.immuni.2018.06.006>.
64. Lenos, K.J., Bach, S., Ferreira Moreno, L., Ten Hoorn, S., Sluiter, N.R., Bootsma, S., Vieira Braga, F.A., Nijman, L.E., van den Bosch, T., Miedema, D.M., et al. (2022). Molecular characterization of colorectal cancer related peritoneal metastatic disease. *Nat. Commun.* 13, 4443. <https://doi.org/10.1038/s41467-022-32198-z>.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti-CD11b Antibody (FITC)	BioLegend	Cat# 101206; RRID: AB_312789
Anti-CD19 Antibody	Cell Signaling	Cat# 90176; RRID: AB_2800152
Anti-CD3e Antibody	Thermo Fisher Scientific	Cat# PA1-29547; RRID: AB_1956724
Anti-CD45 Antibody	Cell Signaling	Cat# 13917; RRID: AB_2750898
Anti-CD8 Alpha Antibody	Abcam	Cat# ab93278; RRID: AB_10563532
Anti-CD8 Alpha Antibody	Cell Signaling	Cat# 98941; RRID: AB_2756376
Anti-EpCAM Antibody (FITC)	BioLegend	Cat# 118208; RRID: AB_1134107
Anti-FOXP3 Antibody	R&D Systems	Cat# MAB8214; RRID: AB_2929004
Anti-HLA-DRA Antibody	Cell Signaling	Cat# 97971; RRID: AB_3714697
Anti-I-A/I-E Antibody (Brilliant Violet 421)	BioLegend	Cat# 107632; RRID: AB_2650896
Anti-PDGFR α Antibody	Cell Signaling	Cat# 3174; RRID: AB_2162345
Anti-PDGFR α Antibody (Alexa Fluor 488)	R&D Systems	Cat# FAB1062G; RRID: AB_3645435
Anti-SPINK4 Antibody	Thermo Fisher Scientific	Cat# PA5-52414; RRID: AB_2647860
Cytokeratin Pan Antibody Cocktail	Thermo Fisher Scientific	Cat# MA5-13203; RRID: AB_10942225
Anti-SPP1 Antibody	Bio X Cell	Cat# BE0382; RRID: AB_2927519
Mouse IgG1 Isotype Control	Bio X Cell	Cat# BE0083; RRID: AB_1107784
Bacterial and virus strains		
pLV-eGFP	Addgene	Cat# 36083
Chemicals, peptides, and recombinant proteins		
Collagenase, Type 1	Worthington Biochemical	Cat# NC9068181
Collagenase, Type 2	Worthington Biochemical	Cat# NC0091709
Collagenase, Type 3	Worthington Biochemical	Cat# NC9405360
Collagenase, Type IV	Thermo Fisher Scientific	Cat# 17104019
Collagenase P	MilliporeSigma	Cat# 11213857001
Elastase	Worthington Biochemical	Cat# NC9030467
Hyaluronidase	MilliporeSigma	Cat# HX0514-1
Deoxyribonuclease I	MilliporeSigma	Cat# DN25
Gey's balanced salt solution	MilliporeSigma	Cat# G9779
Histodenz	MilliporeSigma	Cat# D2158
5-Fluorouracil	MedChemExpress	Cat# HY-90006
Oxaliplatin	MedChemExpress	Cat# HY-17371
Irinotecan	MedChemExpress	Cat# HY-16562
Dead Cell Removal Kit	Miltenyi Biotec	Cat# 130-090-101
CD11b MicroBeads UltraPure, mouse	Miltenyi Biotec	Cat# 130-126-725
CD140a (PDGFR α) MicroBead Kit, mouse	Miltenyi Biotec	Cat# 130-101-502
CD326 (EpCAM) MicroBeads, mouse	Miltenyi Biotec	Cat# 130-105-958
Anti-MHC Class II MicroBeads, mouse	Miltenyi Biotec	Cat# 130-052-401
CellTrace CFSE Cell Proliferation Kit	Thermo Fisher Scientific	Cat# C34554
ImmPRESS HRP Goat Anti-Rabbit IgG Polymer Detection Kit	Vector Laboratories	Cat# MP-7451
ImmPRESS ® HRP Goat Anti-Mouse IgG Polymer Detection Kit	Vector Laboratories	Cat# MP-7452
ImmPRESS AP Horse Anti-Rabbit IgG Polymer Detection Kit	Vector Laboratories	Cat# MP-5401
Opal 4-Color Manual IHC Kit	Akoya Biosciences	Cat# NEL810001KT

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
RNeasy Mini Kit	Qiagen	Cat# 74106
Mouse & Rat Osteopontin ELISA Kit	R&D Systems	Cat# MOST00
Recombinant Mouse Osteopontin/OPN Protein	R&D Systems	Cat# 441-OP
TGF-beta 1 Recombinant Protein	Thermo Fisher Scientific	Cat# 100-21
BD Matrigel matrix, growth factor reduced	BD Biosciences	Cat# 354230
MISSION siRNA Universal Negative Control #1	MilliporeSigma	Cat# SIC001
PDSIRNA2D SASI_Mm01_00052292_ mouse spp1 siRNA	MilliporeSigma	Cat# SASI_Mm01_00052292
PDSIRNA2D SASI_Mm01_00052293_ mouse spp1 siRNA	MilliporeSigma	Cat# SASI_Mm01_00052293

Experimental models: Cell lines

Mouse Cell Line: BMFA3	Huang et al. ⁶²	N/A
Mouse Cell Line: CT1BA5	Huang et al. ⁶²	N/A
Mouse Cell Line: 6620c1	Kerafast	Cat# EUP016-FP
Mouse Cell Line: MC38	MilliporeSigma	Cat# SCC172
Mouse Cell Line: PanMeso	Huang et al. ¹⁶	N/A
Mouse Cell Line: OmMeso	This study	N/A

Experimental models: Organisms/strains

Mouse: C57BL/6J	The Jackson Laboratory	Cat# 000664
Mouse: B6.129S6(Cg)-Spp1tm1Blh/J	The Jackson Laboratory	Cat# 004936
Mouse: Immortomouse (Tg(H2-K1-tsA58)6Kio/LicrmJ	The Jackson Laboratory	Cat# 032619

Deposited data

Public human scRNA-seq datasets	Other public studies	See Table S1 for details
RNA-seq data of PanMeso cells	Huang et al. ¹⁶	GEO: GSE196740
Xenium data of human PM	This study	GEO: GSE274623
Xenium data of human PDAC	This study	GEO: GSE274673
scRNA-seq datasets of <i>Spp1</i> WT and KO mouse pancreatic tumors	This study	GEO: GSE274609

Software and algorithms

CellRanger V7.2.0	10x Genomics	https://www.10xgenomics.com/software
Xenium Ranger V2.0.0.10	10x Genomics	https://www.10xgenomics.com/support/software/xenium-ranger/downloads
Xenium Explorer V3.0.0	10x Genomics	https://www.10xgenomics.com/support/software/xenium-explorer/downloads
R V4.3.2	The R Foundation	r-project.org
Seurat V5.0.1	CRAN	https://cran.r-project.org/web/packages/Seurat/index.html
harmony V1.2.0	CRAN	https://cran.r-project.org/web/packages/harmony/index.html
psych V2.4.3	CRAN	https://cran.r-project.org/web/packages/psych/index.html
tidyverse V2.0.0	CRAN	https://cloud.r-project.org/web/packages/tidyverse/index.html
ggplot2 V3.5.0	CRAN	https://cran.r-project.org/web/packages/ggplot2/index.html
AUCCell V1.24.0	Bioconductor	https://bioconductor.org/packages/release/bioc/html/AUCCell.html
ComplexHeatmap V2.18.0	Bioconductor	https://bioconductor.org/packages/release/bioc/html/ComplexHeatmap.html
GSVA V1.40.1	Bioconductor	https://www.bioconductor.org/packages/GSVA/

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
BPCells V0.1.0	Github	https://github.com/bnprks/BPCells
Monocle3 V1.3.4	Github	https://github.com/cole-trapnell-lab/monocle3
DoubletFinder V2.0.3	Github	https://github.com/chris-mcginnis-ucsf/DoubletFinder
CellPhoneDB V2.1.4	Github	https://github.com/ventolab/CellphoneDB
squidpy V1.2.2	Github	github.com/scverse/squidpy
Python V3.10	Python Software Foundation	python.org
pyscenic V0.12.1	Pypi	https://pypi.org/project/pyscenic
matplotlib V3.9.1	Pypi	https://pypi.org/project/matplotlib
numpy V2.0.0	Pypi	https://pypi.org/project/numpy
pandas V2.2.2	Pypi	https://pypi.org/project/pandas
CytoTRACE 0.3.3	Gulati et al. ⁴²	https://cytotrace.stanford.edu

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Cell lines

BMFA3 and CT1BA5 were mouse primary pancreatic cancer cell lines derived from *Kras*^{LSL-G12D/+}; *Trp53*^{fl/fl}; *Pdx1*^{Cre/+} (KPfC) mice on a C57BL/6 background and isolated as described previously.⁶² *Kras*^{LSL-G12D/+}; *Trp53*^{LSL-R172H/+}; *Ptf1a*^{Cre/+} (KPC) cell clone 6620c1 on a C57BL/6 background was purchased (Kerafast) and has been published in a previous study.⁶³ MC38 murine colon adenocarcinoma cells on a C57BL/6 background were purchased (MilliporeSigma). PanMeso and OmMeso cells were established as follows. Mesothelium was harvested from normal pancreata or omentum of immortomice under the dissecting microscope. The pancreatic or omental mesothelium was seeded onto a tissue culture dish and cultured at 33°C for 20 days. Once cells became confluent, omental mesothelial cells were collected and named omental mesothelial (OmMeso) cells while pancreatic mesothelial cells were subjected to cell sorting. Podoplanin⁺MHC II⁺ cells were collected and named pancreatic mesothelial (PanMeso) cells. All mouse cancer cell lines were cultured in DMEM (Invitrogen) containing 10% FBS and 1% penicillin/streptomycin (Gibco/Thermo) and maintained at 37°C in 5% CO₂. The PanMeso and OmMeso cells were cultured in the mesothelial cell media (medium 199 (Gibco/Thermo), 10% FBS, 1% penicillin/streptomycin (Gibco/Thermo), 400 nM hydrocortisone (MilliporeSigma), 870 nM zinc-free bovine insulin (MilliporeSigma), 20 mM HEPES (Thermo Fisher Scientific)), maintained at 33°C, washed with PBS three times and cultured in new mesothelial cell media at 37°C before any experiments. All cell lines in this study were confirmed to be free of mycoplasma (e-Myco kit, Boca Scientific) before use.

Animals

6-week-old male and female mice were purchased from The Jackson Laboratory, including C57BL/6J mice (JAX stock #000664), *Spp1* knockout mice (JAX stock #004936) and immortomice (JAX stock #032619). All animals were housed in a pathogen-free facility with 24-hr access to food and water. Animal experiments in this study were approved by and performed in accordance with the institutional animal care and use committee at the UTSW Medical Center at Dallas. Mice were euthanized by cervical dislocation under anesthesia.

Patient samples

Formalin-fixed paraffin-embedded (FFPE) tissue sections of T4 colon cancer, PM (from patients who had undergone cytoreductive surgery for colon cancer-derived PM), PDAC (treatment naïve and chemoradiotherapy treated) were obtained with voluntary patient consent from UTSW Tissue Management Shared Resource, which were obtained under a protocol approved by the UTSW Institutional Review Board. All specimens were reviewed by a board-certified pathologist specializing in gastrointestinal pathology to confirm diagnosis. FFPE blocks were cut into 5-μm-thick serial sections. One section was stained with hematoxylin and eosin for pathology review. Tissue sections were used for IHC or spatial transcriptomics in this study. Age and gender of the patients were undisclosed.

METHOD DETAILS

Plasmids

pLV-eGFP (Addgene #36083) was used to expressed GFP in PanMeso cells.

Immunohistochemistry

FFPE tissues were cut in 5- μ m sections. Sections were evaluated by single chromogen (DAB) or multiplex immunohistochemical analysis using antibodies specific for SPINK4 (Thermo Fisher Scientific, 1:500), CD3 (Thermo Fisher Scientific, 1:300), CD8 (for human, Abcam, 1:300; for mouse, Cell Signaling, 1:300), CD19 (Cell Signaling, 1:300), PDGFR α (Cell Signaling, 1:100), HLA-DRA (Cell Signaling, 1:300), CD45 (Cell Signaling, 1:300), FOXP3 (R&D Systems, 1:500), pan-cytokeratin (Thermo Fisher Scientific, 1:500). For DAB staining, tissue sections were deparaffinized in xylene and rehydrated through a graded series of ethanol solutions. Heat-induced epitope retrieval was performed using Tris/EDTA buffer (pH 9.0) in a steamer at 95°C for 30 minutes. Slides were then blocked by 2.5% goat serum for 1 hour and incubated with primary antibodies overnight at 4°C. Following primary antibody incubation, sections were treated with HRP-conjugated secondary antibody and developed using DAB substrate. Nuclei were counterstained with hematoxylin. For multiplex immunofluorescence staining, we used the Opal 4-Color Manual IHC Kit (Akoya Biosciences) following the manufacturer's protocol. Briefly, 5- μ m thick FFPE tissue sections were deparaffinized in xylene and rehydrated through graded ethanol solutions. Antigen retrieval was performed using Tris/EDTA buffer (pH 9.0) in a steamer for 30 minutes at 95°C. Sections were blocked with 2.5% goat serum for 1 hour at room temperature. Primary antibodies were applied sequentially in individual rounds of staining. For each round, sections were incubated with one primary antibody overnight at 4°C, followed by incubation with HRP-conjugated secondary antibody for 1 hour at room temperature. Signal detection was achieved using Opal fluorophores (Opal 520, Opal 570 and Opal 690) diluted 1:250 in amplification diluent. Standard incubation time with Opal fluorophores was 10 minutes at room temperature; however, for the staining of F-apCAFs, to avoid signal saturation in lymphocyte-enriched areas, incubation time was reduced to 2 minutes. After each round of detection, heat-mediated antibody stripping was performed using Tris/EDTA buffer in a steamer for 20 minutes at 95°C to remove bound antibodies before proceeding to the next round. Following completion of all staining rounds, nuclei were counterstained with DAPI (1:1000) for 5 minutes at room temperature. Sections were mounted with ProLong Gold antifade mountant and stored at 4°C in the dark until imaging. For brightfield images, slides were scanned using NanoZoomer 2.0-HT digital slide scanner (Hamamatsu) and images were obtained with NDP.view2 software (Hamamatsu) and analyzed by ImageJ. For fluorescent images, images were obtained with ECHO Revolve microscope and analyzed by ImageJ.

RT-PCR

RNA was extracted using RNeasy Mini Kit (Qiagen). cDNAs were synthesized using the iScript cDNA synthesis kit (Bio-Rad). The expression of genes was measured by qRT-PCR with SYBR-Green Master Mix (Bio-Rad).

Western blot analysis

Cells were lysed using a radioimmunoprecipitation assay buffer (50 mM Tris-HCl, pH 8.0, 150 mM NaCl, 0.1% SDS, 0.5% sodium deoxycholate, and 1% Nonidet P-40) for SDS-PAGE. Protein concentrations were measured with a BCA Protein Assay Kit (Thermo Fisher Scientific). Laemmli sample buffer was added to the lysates, which were then boiled for 10 minutes. Proteins were separated by SDS-PAGE, transferred to nitrocellulose membranes, and blocked with 5% nonfat dry milk or bovine serum albumin. After blocking, membranes were incubated for 1 hour with primary antibodies against SPP1 (BioXCell, 1:1000) in TBST (10 mM Tris-HCl, pH 7.5, 150 mM NaCl, 0.05% Tween 20). The membranes were then washed three times for 10 minutes each with TBST, followed by a 1-hour incubation with horseradish peroxidase-conjugated anti-mouse or rabbit secondary antibodies (Jackson ImmunoResearch Laboratories). After secondary antibody removal with three 10-minute TBST washes, the membranes were treated with SuperSignal West Pico substrate (Thermo Fisher Scientific) to visualize the immunoreactive bands.

Tumor conditioned medium

Mouse PDAC organoids were obtained from *KPC* tumors. The tumor tissues were finely chopped and incubated at 37°C for 12 hours in a digestion buffer, which included 0.012% collagenase XI (MilliporeSigma) and 0.012% dispase (Gibco) in DMEM (Invitrogen) with 1% FBS. The resulting organoids were then pelleted, rinsed with PBS, and plated on a Petri dish in DMEM (Invitrogen) with 10% FBS. After a 4-day culture period of PDAC organoids, MC38 or 6620c1 cancer cells, the medium was passed through a 0.22 μ m ultra-low protein binding filter and collected as PDAC tumor, MC38 colon cancer or 6620c1 PDAC conditioned medium.

Wound healing assay

For the wound healing assay, cells were plated onto 6-well tissue culture plates coated with 50 μ g/ml Matrigel (BD Biosciences) with or without 100 ng/ml recombinant mouse SPP1 protein (R&D Systems) or 1 μ g/ml SPP1 monoclonal antibody (BioXCell). After the cell monolayer reached confluency, the monolayer was scraped with a 200 μ l tip to generate scratch wounds. The cells were washed with PBS and then cultured in medium containing 10% FBS in a tissue culture incubator. At the designated time points after scratch wounds, plates were washed with PBS, and the wound width was photographed and measured under ECHO Revolve microscope.

Isolation of cancer cells, myeloid cells, apCAFs and other CAFs

PM nodules or orthotopic PDAC tumors were first established with MC38 or 6620c1 cancer cells in *Spp1* wildtype or knockout C57BL/6 mice. After one month, PM nodules or PDAC tumors were harvested and single cells were isolated from tumor tissues. Initially, a 10x digestion buffer was prepared in PBS, containing collagenase type I (450 units/ml, Worthington Biochemical), collagenase type II (150 units/ml, Worthington Biochemical), collagenase type III (450 units/ml, Worthington Biochemical), collagenase type IV (450 units/ml, Thermo Fisher Scientific), elastase (0.8 units/ml, Worthington Biochemical), hyaluronidase (300 units/ml,

MilliporeSigma), and DNase type I (250 units/ml, MilliporeSigma). Freshly dissected tumor tissues were cut into 1–2 mm³ pieces using a sterile blade, washed with PBS, and then incubated in 50 mL of 1x digestion buffer on a shaker at 37°C for 45 minutes. The resulting cell suspension was filtered through a 40 µm mesh (MilliporeSigma) and centrifuged at 400 g for 5 minutes. Dead cells were removed with dead cell removal kit (Miltenyi Biotec). Live cells were subjected to sequential MACS separation using MicroBeads following manufacturer instructions (Miltenyi Biotec). The isolation strategy employed a stepwise approach: first, cancer cells were isolated using EpCAM MicroBeads (EpCAM⁺ cells); second, myeloid cells were depleted from the remaining cell suspension using CD11b MicroBeads (CD11b⁺ cells); third, fibroblasts were isolated from the EpCAM⁺CD11b⁺ population using PDGFRα MicroBeads (PDGFRα⁺ cells). Finally, the PDGFRα⁺ fibroblast population was further separated using MHC II MicroBeads into MHC II⁺ (apCAFs) and MHC II[−] (other CAFs) cells. For each separation step, cells were mixed with MicroBeads, incubated for 15 minutes at 4°C, and subjected to MS magnetic column (Miltenyi Biotec) separation. Flow cytometry validation was performed on aliquots from each isolated population to confirm purity. The purity of each cell population was >90% as determined by flow cytometric analysis using the respective surface markers. Each isolation pooled PM nodules or PDAC tumors from at least 10 mice from each group. After MACS separation, each cell type was counted to ensure equal cell numbers were used for subsequent functional experiments.

Colony formation assay

MC38 or 6620c1 cancer cells were cultured in 6-well tissue culture plates at low density (500 cells/well) in 2 ml DMEM (Invitrogen) with 10% FBS. To coculture with apCAFs or other CAFs, we plated 1 × 10⁴ apCAFs or other CAFs into cell culture inserts (pore size 0.4 µm, Thermo Fisher Scientific) with 0.5 ml DMEM with 10% FBS. For SPP1 monoclonal antibody (BioXCell) treatment, the concentration of antibody was 1 µg/ml. Cells were cultured for 12 days when marked colonies formed. Cells were fixed with 10% formalin and stained with crystal violet. Colonies were then counted in each well.

Matrigel invasion assay

For the Matrigel invasion assay, transwells were precoated with 100 ng/ml recombinant mouse SPP1 protein (R&D Systems) or 1 µg/ml SPP1 monoclonal antibody (BioXCell), with 150 µg/cm² Matrigel (BD Biosciences). 5 × 10⁴ MC38 cells were added to the transwells and allowed to migrate towards medium with 10% FBS. After 24 hours, the cells on the top side of the transwells were scraped off with cotton swabs. The transwell membrane was then fixed with 4% paraformaldehyde for 20 minutes. Cells were stained with crystal violet and counted under microscopy at 100× magnification. For apCAF/other CAF-facilitated Matrigel invasion assay, 1 × 10⁴ MC38 cells were first labeled with CFSE (Invitrogen), mixed with 1 × 10⁴ apCAFs or other CAFs isolated from *Spp1* wildtype or knockout PM nodules and plated to the Matrigel-coated (+/-SPP1 antibody) transwells. After 24 hours, the cells on the top side of the transwells were scraped off with cotton swabs. CFSE⁺ cells were counted under ECHO Revolve microscope.

Chemosensitivity assay

6620c1 cancer cells were labeled with CFSE (Invitrogen) following the manufacturer protocol, mixed with PanMeso cells or apCAFs at a 1:1 ratio and seeded at a density of 6,000 cells per well in 100 µL of DMEM (Invitrogen) with 10% FBS. To examine the effect of SPP1 protein or antibody, 100 ng/ml recombinant mouse SPP1 protein (R&D Systems) or 1 µg/ml SPP1 monoclonal antibody (BioXCell) was used. The next day, various concentration (0–20 µM) of FOLFIRINOX (5-FU: Oxaliplatin: Irinotecan=1:1:1, MedchemExpress) was added to the wells. Cells were incubated for 36 hours and measured at excitation: 492 nm/emission: 517 nm using a microplate reader. To knockdown *Spp1*, PanMeso cells were transfected with control siRNA (MISSION siRNA universal negative control, MilliporeSigma) or *Spp1* siRNA (SASI_Mm01_00052292 or SASI_Mm01_00052293, MilliporeSigma) one day before seeding for coculture assay. Effect of knockdown was validated by western blot against SPP1.

SPP1 ELISA

To determine SPP1 secretion from different cell populations, we collected conditioned media from MC38 cancer cells, 6620c1 cancer cells, myeloid cells, apCAFs, and other CAFs. Cells were cultured for 96 hours in serum-free DMEM before media collection. Secreted SPP1 levels were measured using a mouse SPP1 Quantikine ELISA Kit (R&D Systems) according to the manufacturer's protocol. Briefly, samples and standards were added to microplates pre-coated with mouse SPP1-specific antibody, followed by incubation with enzyme-linked polyclonal antibody specific for mouse SPP1. After washing, substrate solution was added to produce a colorimetric reaction proportional to SPP1 concentration. Optical density was measured at 450nm with wavelength correction at 540nm. SPP1 concentrations were calculated using a standard curve generated with recombinant mouse SPP1.

PSC isolation and coculture

To isolate PSCs, pancreata from C57BL/6 mice were finely minced and subjected to digestion at 37°C for 30 minutes using a dissociation buffer containing 0.05% collagenase P (MilliporeSigma) and 0.1% DNase I (MilliporeSigma) in Grey's balanced salt solution (GBSS; MilliporeSigma). Following digestion, the tissue was filtered through a 100-µm nylon mesh and subsequently washed in GBSS with 0.3% BSA and 0.1% DNase I. After centrifuging, the cell pellet was resuspended in 9.5 ml of GBSS containing 0.3% BSA and 43.75% Histodenz (MilliporeSigma). 6 ml of GBSS with 0.3% BSA was carefully placed on top of the cell suspension, and the gradient was subjected to centrifugation for 20 minutes at 1,400 RCF, with the brake off. Cells located in the fuzzy band just above the interface between Histodenz and GBSS were collected, washed with PBS, and plated into 6-well plate (2 × 10⁵ cells per well). To coculture with apCAFs, we plated 1 × 10⁴ apCAFs into cell culture inserts (pore size 0.4 µm, Thermo Fisher Scientific). To

induce myCAF formation, TGF β 1 (30 ng/mL, Thermo Fisher Scientific) was added to the PSCs with or without apCAFs. To examine the effect of SPP1 antibody, 1 μ g/ml SPP1 monoclonal antibody (BioXCell) was added to the coculture. PSCs and apCAFs were cocultured for 24 hours and PSCs were harvested for RT-PCR to detect gene changes.

In vivo models

For PM, MC38 cells were injected intraperitoneally (5×10^4 cells) in 6-week-old C57BL/6 mice (male and female). Mice were sacrificed 1 month after injection and PCI score was calculated based on published literature.⁶⁴ For PDAC, 6620c1 cells were injected orthotopically (2×10^5 cells) in 6-week-old C57BL/6 mice (male and female). To harvest tumors, mice were sacrificed 1 month after injection. Tumors were then subjected to digestion, scRNA-seq or IHC staining. For therapies, C57BL/6 mice were treated with control (clone MOPC-21, BioXCell, 500 μ g/dose) or anti-SPP1 Ab (MPIIB10, BioXCell, 500 μ g/dose) 7 days after cancer cell injection and maintained two doses per week during the tumor formation.

Single cell cDNA library preparation and sequencing

Single cells were isolated from mouse tumor tissues. 10x digestion buffer was first prepared in PBS, containing collagenase type I (450 units/ml, Worthington Biochemical), collagenase type II (150 units/ml, Worthington Biochemical), collagenase type III (450 units/ml, Worthington Biochemical), collagenase type IV (450 units/ml, Thermo Fisher Scientific), elastase (0.8 units/ml, Worthington Biochemical), hyaluronidase (300 units/ml, MilliporeSigma), and DNase type I (250 units/ml, MilliporeSigma). Freshly dissected tumor tissues were cut into 1–2 mm³ pieces using a sterile blade, washed with PBS, and then incubated in 50 mL of 1x digestion buffer on a shaker at 37°C for 45 minutes. The resulting cell suspension was filtered through a 40 μ m mesh (MilliporeSigma) and centrifuged at 400 g for 5 minutes. After removing the supernatant, the cell pellet was resuspended in ACK lysing buffer (Thermo Fisher Scientific) and incubated on ice for 5 minutes to lyse red blood cells.

For each group, we included 6 tumors. Every two tumors were then pooled together for library construction (three libraries per group). Approximately 10,000 cells were used for library construction with the Chromium Next GEM Single Cell 3' Kit v3.1 (10x Genomics), following the manufacturer's protocol. The purified libraries were sequenced on an Illumina NextSeq6000. Raw base call (BCL) files from the sequencer were demultiplexed into fastq files. These fastq files were aligned to the mm10 mouse reference genome and quantified using the Cell Ranger pipeline (version 7.2.0, 10x Genomics). The preliminary filtered data generated from Cell Ranger were utilized for all subsequent analyses.

Single-cell RNA-seq data processing

The count matrices generated by Cell Ranger were imported using the Read10X function from the Seurat package (v5.0.1) and converted into Seurat objects. Quality control was performed based on the number of detected genes and the proportion of mitochondrial gene counts per cell. Specifically, cells with fewer than 200 detected genes or more than 6000 detected genes, as well as cells with a mitochondrial unique molecular identifier (UMI) count percentage greater than 15%, were excluded. Additionally, genes detected in fewer than 3 cells were removed. DoubletFinder (v2.0.3) with default settings was used to identify and exclude potential doublets.

Normalization was carried out using the NormalizeData function to ensure uniform total gene expression per cell, with the scaling factor set to 10,000. The FindVariableFeatures function identified the top 2000 highly variable genes (HVGs). Seurat objects were then scaled with the ScaleData function before performing principal component analysis (RunPCA), retaining the first 50 principal components (PCs) for further analysis. Seurat objects from multiple samples were merged using the merge function. To correct for batch effects, the Harmony algorithm (v1.2.0) was applied with default parameters (RunHarmony, max.iter.harmony = 10) using sample labels as the covariates.

A shared nearest neighbor (SNN) graph was constructed using the FindNeighbors and FindClusters functions in Seurat, leveraging the appropriate PCA dimensions. Cells were clustered using the Louvain algorithm in a graph-based approach. The RunUMAP function (reduction = "harmony") was used to generate the UMAP embedding. Marker genes and differentially expressed genes for each cluster were identified using the FindAllMarkers and FindMarkers functions in Seurat. Cell classification into major lineages was guided by literature marker genes. Visualization was achieved using the DimPlot, FeaturePlot, and VlnPlot functions.

Subclustering analysis followed a similar methodology, including normalization, identification of variable features, dimensionality reduction with Harmony-based batch correction, and cluster identification.

The identification of major cell types relies on the expression patterns of canonical marker genes, including epithelial cells (*EPCAM*, *CDH1*, *KRT18*), T & NK cells (*CD3D*, *CD3E*, *CD7*), myeloid cells (*AIF1*, *CD14*, *CD68*), endothelial cells (*PECAM1*, *VWF*, *CDH5*), B cells (*CD19*, *CD79A*, *CD79B*), plasma cells (*CD38*, *JCHAIN*, *IGKC*), mast cells (*TPSB2*, *TPSAB1*, *MS4A2*), and fibroblasts (*COL1A1*, *COL1A2*, *COL3A1*). Clusters exhibiting high expression of classical fibroblast marker genes (*COL1A1*, *COL1A2*, *COL3A1*) were selected for subsequent CAF subpopulation analysis.

Cell cluster proportion and hierarchical clustering of different cancer types

To quantify the distribution of cell clusters across various cancer types, we calculated the proportion of each cell cluster within each cancer type. Additionally, we performed unsupervised hierarchical clustering for all cancer types based on these cluster proportions. The frequencies of all clusters served as input for the hierarchical clustering, which was carried out using the hclust function.

Calculation of signature score

To assess the functional differences among CAF subpopulations, we identified gene sets corresponding to 10 specific gene signatures (Table S3). The AUCell package (v1.24.0) in R was used to compute the signature scores for these gene sets. Initially, the AUCell_buildRankings function was utilized to create the ranking expression matrix. Next, the AUC values were calculated using the AUCell_calcAUC function. Visualization of the results was accomplished with the ggplot2 package in R.

Similarity analysis of CAF subclusters

To measure the similarity between different CAF subclusters, we utilized the AverageExpression function from the Seurat package to compute the average expression values for all genes across the subclusters. The correlation coefficients between the subclusters were then calculated using the corr.test function from the psych package (v2.4.3). Visualization was performed with the pheatmap package (v1.0.12).

Enrichment of fibroblast subclusters in normal and tumor tissues

To quantify the enrichment of each fibroblast subcluster in normal and tumor tissues, we initially grouped all cells by tissue types and cell subclusters to determine the total number of cells in each group, converting this information into a matrix. We then computed the marginal tables for rows, columns, and the overall dataset, followed by normalizing the data based on these marginal tables. Tissue types with normalized values greater than 1 were categorized as enrichment, while those with values less than 1 were categorized as depletion.

Pseudotime trajectory inference

Monocle3 (v2.30.1) was used to infer cell type transition states among apCAF subclusters. Dimensionally-reduced data from the integrated Seurat object was converted into Monocle3 cell_data_set objects. To facilitate the learning of disjoint graphs across multiple partitions, cells were re-clustered. A trajectory graph was then constructed, accommodating closed loops and multiple partitions. For each compartment, the starting point of the trajectory was determined as the node closest to the previously identified progenitor populations. Pseudotime along the trajectory was calculated using this designated starting node.

Gene regulatory network inference

The SCENIC algorithm in Python (pySCENIC v0.12.1) was used to infer regulatory interactions between transcription factors and their target genes. The process began by constructing a gene co-expression network using the grn function with the GRNBoost2 algorithm. Regulons were subsequently refined based on the hg38 cistarget database. The aucell command was then employed to score regulon activity in each cell. Differentially activated regulons within each CAF subset were identified using the wilcox.test with BH-adjusted p-values < 0.05.

Cellular abundance analysis of apCAF

To investigate the abundance and distribution of apCAFs across different cancers, we performed a comprehensive cellular composition analysis for quantifying and visualizing cell type distribution. Our methodology comprised of three critical steps: First, we generated a contingency table of cell type frequencies using dplyr's grouping functions and replacing missing values with zero. Second, we performed an enrichment analysis by calculating marginal totals, computing expected cell count matrices, and deriving enrichment ratios through comparison of observed versus expected cell counts. Third, we employed ggpubr's ggdotchart for statistical visualization. Enrichment was defined as values > 1 (over-representation), with a horizontal baseline at y=1 to facilitate comparative assessment of apCAF abundance across different cancers. The result was visualized using a dot plot, where each dot represents the proportion of apCAFs for a specific cancer type.

Cell-cell interaction analysis

The Python package CellPhoneDB (v3.0) was utilized to infer interactions between apCAFs and immune cells/cancer cells. Input files for the statistical analysis, including the raw count matrix and the cell type annotation files, were extracted from the Seurat object we created. Enriched ligand-receptor pairs between cell subclusters were identified by performing a permutation test on the expression matrix. Significant ligand-receptor pairs with a p-value < 0.05 were selected for illustration.

Survival analysis

We separated apCAFs into normal and tumor tissues for differential expression analysis within the two apCAF populations. Genes up-regulated in cancer were identified, and the specificity of each gene was assessed by evaluating whether it was distinctly expressed in each apCAF population compared to other CAF subtypes. This analysis led to the identification of three genes most specific to F-apCAFs (*C1QC*, *APOC1*, *RARRES1*) and M-apCAFs (*CD24*, *CA9*, *EGLN3*) as cancer-related signatures for the two apCAF populations. Survival analysis was subsequently conducted using the GEPIA2 database, based on the expression levels of the multi-gene signature for each apCAF population, and Kaplan-Meier curves were generated to visualize the combined survival of TCGA solid tumor types (all tumor types available on GEPIA2).

Enrichment analysis

Pathway enrichment analysis was conducted with Metascape (<https://metascape.org>) or GSEA, and cell-type specific enrichment analysis was performed using WebCSEA (<https://bioinfo.uth.edu/webcsea/>).

CytoTRACE analysis

The CytoTRACE package was used to evaluate the transcriptional diversity of each cell, focusing on their stem cell or differentiation status. Each cell was assigned a CytoTRACE score, where higher scores corresponded to greater stemness and less differentiation characteristics.

GeoMx spatial profiling

Nanostring GeoMx Human Whole Transcriptome Atlas (WTA) platform was used to profile a total of 96 ROIs from 8 PM patient samples (12 per sample). This panel profiles the whole transcriptome by targeting over 18,000 unique transcripts from human protein encoding genes plus ERCC negative controls. The panel excludes uninformative high-abundance RNAs such as ribosomal subunits and includes RNA probes designed for Illumina NGS readout with the Seq Code library prep. Tissue slides were incubated with the fluorophore-conjugated antibodies against vimentin (Santa Cruz, clone E-5), pan-cytokeratin, and a cocktail of photocleavable oligonucleotide probes from the GeoMx kit. A board-certified pathologist selected ROIs based on histology and morphology marker staining. Libraries were prepared according to the GeoMx manual and sequencing depth was 100 read pairs/ μm^2 .

Process sequenced FASTQ files were first converted to DCC (digital count files) run by Nanostring's GeoMx NGS Pipeline (Version 2.3.3.10). DCCs were transferred to the GeoMx DSP Data Analysis Suite. Readout-specific quality control was checked and standardized QC threshold settings was set as recommended by the manufacturer. All 96 ROIs passed the segment quality control. Gene filtering retained 11903 of 18677 (63.73%) genes after removing targets with expression at or lower than selected threshold value at or above specific frequency (5% in our case). Q3 normalization was applied to account for differences in sequencing yield, enabling biological comparison of gene expression across ROIs. The normalized count density distributions across individual ROIs were grossly similar, and there was no observable bias among samples or by histology (Table S4).

GeoMx data analysis

To extract the iCMS2 and iCMS3 epithelial signatures, we downloaded the count expression matrix and metadata for epithelial cells from Synapse Storage (ID: syn26844071). For gene expression counts, individual samples were merged and analyzed using the Seurat 4.3 software package. Cells with UMI counts less than 500 or more than 20% mitochondrial counts were removed. Further filtering of cells in the Seurat object was done by removing normal epithelial cells. Epithelial cancer cell spatial deconvolution for each segmented tumor ROI were performed with the Bioconductor package SpatialDecon based on the epithelial cell profile matrix from the iCMS dataset. Principal component analysis for spatial data of epithelial cells were generated by ClustVis. Samples were grouped into iCMS2 or iCMS3 based on the deconvolution information. Top 500 differential expressing genes of each group were used to complete the analysis (Table S4). Differentially expressed genes (DEGs) of GeoMx RNA data between two iCMS cancer cell or iCMS stroma subgroups were determined using the limma package in R. $|\text{Log2FC}| > 0.75$, and $\text{FDR} < 0.05$ were considered statistically significant. The volcano plots of DEGs were visualized by ggpubr R package. KEGG pathway analysis of DEGs between iCMS3 and iCMS2 stroma was performed using the enricher function from the clusterProfiler R package, with a significance threshold of $\text{qvalueCutoff} = 0.05$. The gene sets employed in this analysis were sourced from the Molecular Signatures Database (MSigDB 2024.1.Hs) (Table S5).

Imaging mass cytometry

5- μm -thick sections were cut from FFPE tissue blocks onto glass slides (3 iCMS2 and 3 iCMS3 PM samples) and stained with a cocktail of metal-conjugated antibodies against immune markers as well as cytoplasmic and nuclear markers provided by the company that were used for cell segmentation (Fluidigm). The tissue sections were then ablated by Hyperion system for data acquisition (Fluidigm). Two images per sample were screened and the MCD files were processed with MCD Viewer. TIFF files were then extracted from MCD files for downstream analyses. For cell segmentation, machine learning was utilized in CellProfiler 4.2.5 for pixel classification of cell nuclei, enabling cell segmentation. CellProfiler generates segmentation masks that, when combined with individual TIFF files, allow for the extraction of single-cell information from each image. Next, histoCAT (high-dimensional cytometry analysis toolbox) was employed to extract single-cell data from the images of the segmentation masks. histoCAT enables the measurement of abundance for all markers of interest within cells and regions of interest. This step helped capture the high-dimensional characteristics of each individual cell. Then, we output standardized data from histoCAT and imported the Seurat4.3 R package for further analysis.

Xenium *in situ* high-plex assay

Based on the single-cell atlas data, we designed a custom gene panel (Table S6) comprising 480 genes for cell type recognition and gene functional analysis. Tissue sections were prepared according to the Xenium *In Situ* for FFPE-Tissue Preparation Guide protocol (10X Genomics). Briefly, FFPE sections were cut to 5 μm and placed in the designated sample area (10.45mm x 22.45mm) on a Xenium slide (10X Genomics). The sections were then dewaxed in xylene, rehydrated through a gradient ethanol series, and treated with enzyme to facilitate mRNA accessibility. Probe hybridization was performed at 50°C overnight, followed by the removal of unhybridized probes. Locked DNA probes bound to target mRNA were ligated at 37°C for 2 hours (Xenium ligase A/B). Circular DNA

probes for each gene underwent rolling circle amplification, producing multiple copies of specific DNA barcodes. After washing, background fluorescence was chemically quenched. The slides were then analyzed using a Xenium analyzer (10X Genomics) for image acquisition. The morphological image of DAPI nuclear staining was used for cell assignment, and built-in machine learning algorithms were applied for cell segmentation. Output files generated by the instrument's pipeline were used for downstream analysis.

Xenium data processing

We followed the Seurat vignette (https://satijalab.org/seurat/articles/seurat5_spatial_vignette_2.html) to load and analyze the Xenium output data using the Seurat package. We read and merged the cell x gene matrices to create a Seurat object. The SCTransform function was used for normalization, followed by standard dimensionality reduction and clustering. The clustering results were visualized in UMAP space using the DimPlot function.

For each Xenium sample, we performed robust cell type decomposition (RCTD) with the R package spacexr (v2.2.1) to deconvolve spatial transcriptomic data into cell types, using the Seurat reference constructed from our pan-cancer scRNA-seq atlas data. The cell type deconvolution results were incorporated into the Seurat object metadata for subsequent analysis. The BuildNicheAssay function was used to compute spatial niches by calculating the cell type composition of each cell using the k-nearest neighbor algorithm. Cells with similar neighborhoods were grouped into spatial niches using k-means clustering. Seurat's ImageDimPlot and ImageFeaturePlot functions were leveraged to visualize transcript localization and segmentation boundaries. Additionally, these visualizations, along with clustering outputs, were produced using the Xenium Explorer 3 software.

To investigate the distribution of various cell types within lymphocyte-enriched niches, we selected 10 lymphocyte-enriched regions from 5 samples for analysis. First, we identified the lymphocyte-enriched regions and quantified the number of cells for each cell type within these regions. The relative proportion of each cell type was then calculated by dividing the cell count of each type by the total number of cells in the lymphocyte-enriched region. These proportions were subsequently visualized using graphical representations. The baseline $y = 0.714$ represented the relative proportion of each cell type and was used to evaluate the enrichment level of each cell type within the lymphocyte-enriched regions.

Spatial distance analysis between F-apCAFs, myCAFs, and lymphocyte-enriched aggregates was conducted in the lymphocyte-enriched regions. T, B, and Plasma cells were classified as lymphocytes. For the F-apCAF or myCAF subset, the distance between a specific cell k in this subset and lymphocyte aggregates was defined as the shortest Euclidean distance d between cell k and all cells j in the lymphocyte-enriched niches, expressed as:

$$D_k = \min_{j \in \text{Lymph}} d_{kj}$$

Finally, we calculated the kernel density estimates of all D_k in a specific subset using the density function and visualized it.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analyses and graphical representations of the data were primarily performed using the R statistical environment (<https://cran.r-project.org>, version 4.3.2) and GraphPad Prism software (version 10.0.0). Relative risks (RR) were computed using bivariate log-binomial regressions, considering different sites of metastasis as the outcome and SPINK4 (+ or -) as the exposure. This analysis was conducted with SAS 9.4 statistical software, with significance determined by a two-sided α level of ≤ 0.05 . The Wilcoxon Rank-Sum test was used to compare means between two groups of continuous variables, while Spearman's rank correlation was employed to assess correlations across multiple groups. The Benjamini-Hochberg method was applied for multiple test corrections, and false discovery rates (FDR) were calculated accordingly. *In vivo* studies were quantified with t test. A P or FDR value of less than 0.05 was deemed statistically significant, with symbols for significance as follows: ns, non-significant; * <0.05 , ** <0.01 , *** <0.001 , **** <0.0001 . Illustrations were created using [BioRender.com](https://www.biorender.com) or Adobe Illustrator software.